Module - 02

Exploratory Data Analysis (EDA)

2.1 Descriptive statistics: mean, median, mode, variance, standard deviation, percentiles using pandas and Scipy, 2.2 Data distributions: normal, skewed, kurtosis; probability distributions, 2.3 Correlation and covariance; introduction to hypothesis testing basics (t-test, chi-square), 2.4 Data visualization techniques: histograms, box plots, scatter plots, heatmaps ,2.5 EDA using Pandas and Matplotlib/Seaborn libraries ,2.6 Univariate, bivariate, and multivariate analysis

“Suppose I give you the marks of 100 students. Can you understand the performance of the class just by looking at all 100 marks?”

For example:

Student	Marks
A	45
B	67
C	78
D	56
E	89
F	72
G	34
H	91
I	65
J	78
...	

“Instead of looking at all 100 marks, what if I tell you the average mark is 68?”

Now we immediately understand something about the class.

Is the average enough?”
No.

We may also want to know:

- What is the middle mark?
- Which mark occurs most often?
- Are students' marks close together or very different?
- What percentage of students scored below a particular mark?

This is where **Exploratory Data Analysis** comes in.

What is Exploratory Data Analysis?

Exploratory Data Analysis (EDA) is the process of examining, summarizing, and understanding a dataset before performing further analysis or building a model. (OR)
EDA means exploring and understanding our data before making decisions or performing further analysis. (OR)
EDA helps us convert a large, difficult-to-understand dataset into meaningful information that we can interpret and use for further analysis.

we have a student dataset:

Student	Gender	Department	Marks	Attendance
A	F	CSE	85	92
B	M	CSE	72	85
C	F	ECE	65	78
D	M	CSE	90	95
E	F	ECE	55	70

Before making conclusions, we need to understand the data.

Why do we need EDA?

Imagine a college has the marks of 500 students.

The principal asks:

“How did students perform in the examination?”

We could give the principal all 500 marks.
But that isn't very useful.

Instead, we can summarize the data:

- Number of students = 500
- Average mark = 68
- Median mark = 70
- Most common mark = 75
- Standard deviation = 10
- 25th percentile = 60
- 75th percentile = 78

Now the principal can understand the overall performance quickly.
EDA helps us convert a large amount of raw data into useful information that is easier to understand.

Main Components of EDA

EDA generally involves:

A. Understanding the dataset

- **df.head()**
- **df.shape**
- **df.info()**
- **df.describe()**

B. Checking data quality

- **df.isnull().sum()**
- **df.duplicated().sum()**

C. Descriptive statistics (Important concept)

- **Mean**
- **Median**
- **Mode**
- **Variance**
- **Standard deviation**
- **Percentiles**

What is Descriptive Statistics?

"Descriptive statistics are numbers used to summarize and describe the important characteristics of data."

(OR)

"Descriptive statistics summarizes a large amount of data using a few meaningful numbers."

For example:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Percentiles

Example: Student Marks

let's assume these are the marks of 10 students."
45, 50, 55, 60, 65, 70, 70, 75, 80, 90

Write them in ascending order:
45
50
55
60
65
70
70
75
80
90

1. Mean — The Average

It answer the question "What is the average mark of the students?"

Definition : Mean is simply the average of all the values.

Mean Formula

$$\text{Mean} = \frac{\text{Sum of All Data Points}}{\text{Number of Data points}}$$

Step 1: Add all marks
45 + 50 + 55 + 60 + 65 + 70 + 70 + 75 + 80 + 90
= 660

Step 2: Count the students
There are 10 students.

Step 3: Divide
Mean = 660 / 10
= 66
So:
The average mark is 66.

Real-Life Example of Mean

Suppose five employees earn:
₹ 20,000
₹ 25,000
₹ 30,000
₹ 35,000
₹ 40,000

Mean salary:
₹ 30,000

We can say:
"The average salary is ₹ 30,000."

So mean is useful when we want a **quick overall average**.

But Mean Has a Problem

Now take this example:

₹ 20,000
₹ 25,000
₹ 30,000
₹ 35,000
₹ 5,00,000

"Does ₹ 5,00,000 look like the other salaries?

No.

It is an **extreme value**.

Calculate the average:

(20,000 + 25,000 + 30,000 + 35,000 + 5,00,000) / 5

= ₹ 1,22,000

"Does ₹ 1,22,000 represent what most employees earn?"

Not really.

Mean can be affected significantly by extreme values.

2. Median — The Middle Value

"If the data is arranged in order, median tells us the middle of the data."

Our data:
45, 50, 55, 60, 65, 70, 70, 75, 80, 90
There are 10 values.

Because there are an **even number of values**, we have two middle values:
45 50 55 60 [65 70] 70 75 80 90
↑ ↑
middle values
Take their average:
Median = (65 + 70) / 2

= 67.5
Therefore:
Median = 67.5

Why is Median Useful?

Return to the salary example:

₹ 20,000
₹ 25,000
₹ 30,000
₹ 35,000
₹ 5,00,000

Arrange them:

20,000
25,000
30,000
35,000
5,00,000

The middle value is:

₹ 30,000

So:

Median = ₹ 30,000

Now compare:

Mean = ₹ 1,22,000
Median = ₹ 30,000

Which better represents the typical employee?

Median.

So,

When data contains extreme values, median can give a better picture of the typical value than mean.

Mean is sensitive to extreme values, while median is less affected by extreme values.

3. Mode — Most Frequently Occurring Value

"Look at our marks. Which mark occurs more than once?"

45, 50, 55, 60, 65, 70, 70, 75, 80, 90

↑

70

70 occurs twice.

Therefore:

Mode = 70

Simple definition

Mode is the value that occurs most frequently in the data.

Real-Life Example of Mode

Suppose a clothing shop records T-shirt sizes sold:
M, L, M, S, M, XL, L, M, S

Count them:
M → 4
L → 2
S → 2
XL → 1

Therefore:
Mode = M

The shop manager can conclude:
"Medium size is the most frequently purchased size."

So, mode is especially useful for **frequently occurring values**.

4. Variance — Measuring Spread

First understand the problem

Suppose there are two classes, and both classes have an average mark of **50**.

Class A

48, 49, 50, 51, 52

Class B

20, 35, 50, 65, 80

Both have:

Mean = 50

But look at them:

Class A: 48 49 50 51 52

↑

Mean

Class B: 20 35 50 65 80

↑

Mean

In Class A, students' marks are **very close to 50**.

In Class B, students' marks are **far away from 50**.

So, we need a number that tells us:

"How spread out are the values around the average?"

That number is called **variance**.

Variance tells us how much the values are spread out from the mean.
Think of the mean as the **center**.
For example:

Mean
↓
20 35 50 65 80
← spread →

If values are very close to the mean:
48 49 50 51 52
↑
mean
the variance is small.

If values are far away:
20 35 50 65 80
↑
mean
the variance is large.

Both classes have the same average, but their performances are very different.

Simple Variance Calculation

Use a very small dataset so you can understand it.

40, 50, 60

Step 1: Find the mean

Mean = (40 + 50 + 60) / 3

= 50

Step 2: Find how far each value is from the mean / finding the difference

Value	Mean	Difference
40	50	-10
50	50	0
60	50	+10

40 - 50 = -10

50 - 50 = 0

60 - 50 = +10

Step 3: Square the differenc

$(-10)^2 = 100$

$(0)^2 = 0$

$(10)^2 = 100$

Step 4: Find the average of the squared differences

Variance = (100 + 0 + 100) / 3

= 66.67

So, the variance is approximately:

66.67 ,this is the variance of 40, 50, 60

Why Do We Square?

Because without squaring:

-10 + 0 + 10 = 0

The negative and positive values cancel each othe

It would incorrectly look like there is no variation.

After squaring:

100 + 0 + 100 = 200

Now the spread is represented properly.

5. Standard Deviation

Why we are using Standard Deviation?

There is a problem with variance.
What problem is:
The original data is in **marks**.
But variance is in **squared marks**.
That is not very easy to understand.

So, we need another measure that tells us the spread in the **same unit as the original data**.
That measure is:
Standard Deviation

Standard deviation tells us approximately how much the values differ or spread from the mean.

(OR)

Standard deviation tells us how spread out the data is around the average.

To get standard deviation, we simply take the **square root of variance**.

Standard Deviation = √ Variance

Therefore:
Standard Deviation = √ 66.67

Approximately:
Standard Deviation = 8.16

Variance and standard deviation both measure spread, but standard deviation is easier to interpret because it is expressed in the same units as the original data.

(OR)

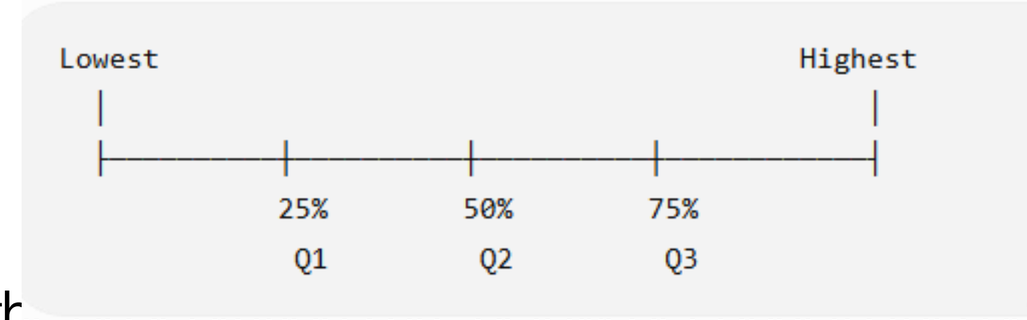
Variance is useful mathematically; standard deviation is easier to understand

6. Percentiles

A percentile tells us the position of a value relative to the other values in the dataset.

25th, 50th and 75th Percentiles

Imagine 100 students arranged from lowest marks to highest marks.



- **25th percentile**

Called **Q1**.

Approximately 25% of the observations are below or equal to this point.

- **50th percentile**

Called **Q2**.

This is the **median**.

Approximately 50% of the observations are below or equal to this point.

- **75th percentile**

Called **Q3**.

Approximately 75% of the observations are below or equal to this point.

Here, how do we actually calculate 25th, 50th, 75th percentile

Take the data: 45, 50, 55, 60, 60, 65, 70, 75, 80, 90

Arrange the data:

Position	1	2	3	4	5	6	7	8	9	10
value	45	50	55	60	60	65	70	75	80	90

Calculate the 25th percentile.

$$\text{position} = \frac{P}{100}(n-1)+1$$

where P = percentile.

n = number of observations.

Here $P = 25$, $n = 10$

$$\therefore \quad \text{Position} = \frac{25}{100}(10-1)+1$$

$$= 0.25(9)+1$$

$$= 2.25+1$$

$$= 3.25$$

So, the 25th percentile lies between the 3rd and 4th value.

3rd value = 55

4th value = 60

Now take 25% of the difference.

$$60 - 55 = 5$$

$$25\% \times 5 = 1.25$$

Add the 1.25 value to the 55 because it is the starting point of the 3rd value.

$$55 + 1.25 = 56.25$$

Therefore

$$P_{25} = 56.25$$

Calculate the 50th percentile.

The 50th percentile is the median.

So, the middle value is

5th value = 60

6th value = 65.

$$P_{50} = \frac{60+65}{2} = \frac{125}{2}$$

$$P_{50} = 62.5$$

Calculate the 75th percentile.

$$\text{Position} = \frac{P}{100}(n-1)+1$$

where P = percentile.
n = number of observations

Here P = 75, n = 10

$$\therefore \quad \text{Position} = \frac{75}{100}(10-1)+1$$

$$= 0.75(9)+1$$

$$= 6.75+1$$

$$= 7.75$$

So, the 75th percentile lies b/w 7th & 8th value.

7th value = 70

8th value = 75

$$75-70=5$$

$$75\% \times 5 = 3.75$$

$$70+3.75 = 73.75$$

Therefore.

$$P_{75} = 73.75$$

Panda’ s quantile()

For 25th percentile:
marks.quantile(0.25)

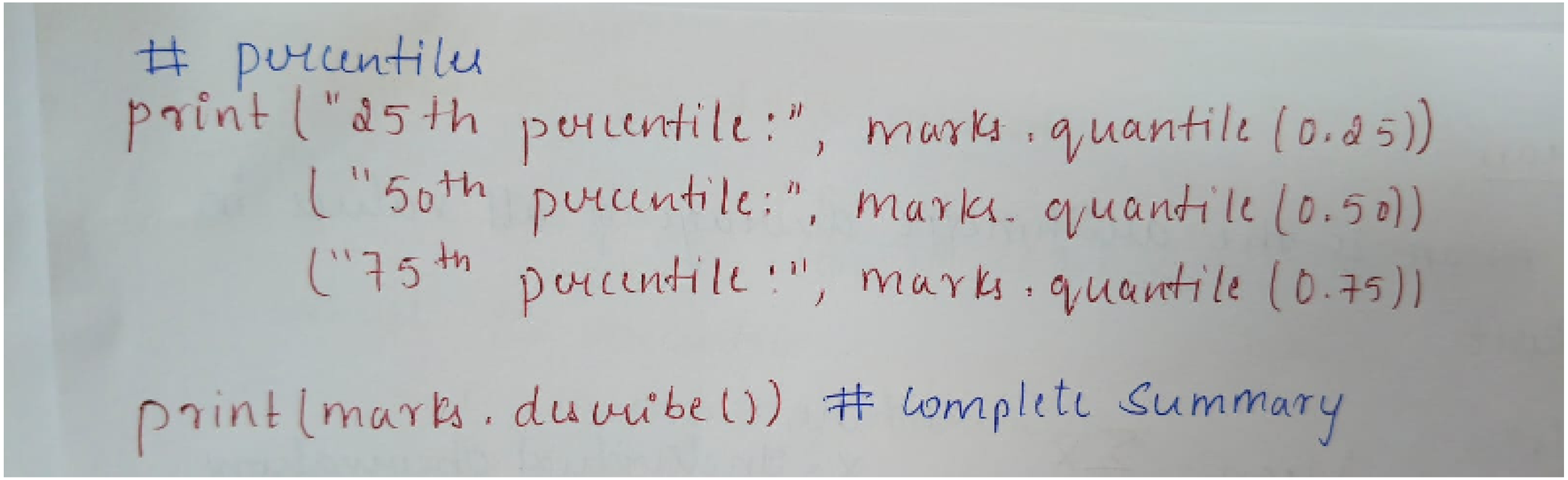
For 50th percentile:
marks.quantile(0.50)

For 75th percentile:
marks.quantile(0.75)

You can calculate all three together:
marks.quantile([0.25, 0.50, 0.75])

Code: using pandas

```
Import pandas as pd
# Create the dataset
marks = pd.Series([45,50,35,60,60,65,70,75,80,90])
# Descriptive statistics
print("Mean:", marks.mean())
      ("Median:", marks.median())
      ("mode:";.marks.mode())
      ("Variance:", marks.var())
      ("Standard Deviation:", marks.std())
# Minimum and Maximum
print("Minimum:", marks.min())
print("Maximum:", marks.max())
```


percentiles
print("25th percentile:", marks.quantile(0.25))
("50th percentile:", marks.quantile(0.50))
("75th percentile:", marks.quantile(0.75))
print(marks.describe()) # complete Summary

“Sometimes we need more advanced mathematical and statistical operations. For that,
Python provides SciPy.”

SciPy library stands for **Scientific Python**.

We import the statistics part of SciPy like this:
from scipy import stats
Now we can use statistical functions from stats.

Code using Scipy

```
Import numpy as np
from scipy import stats.

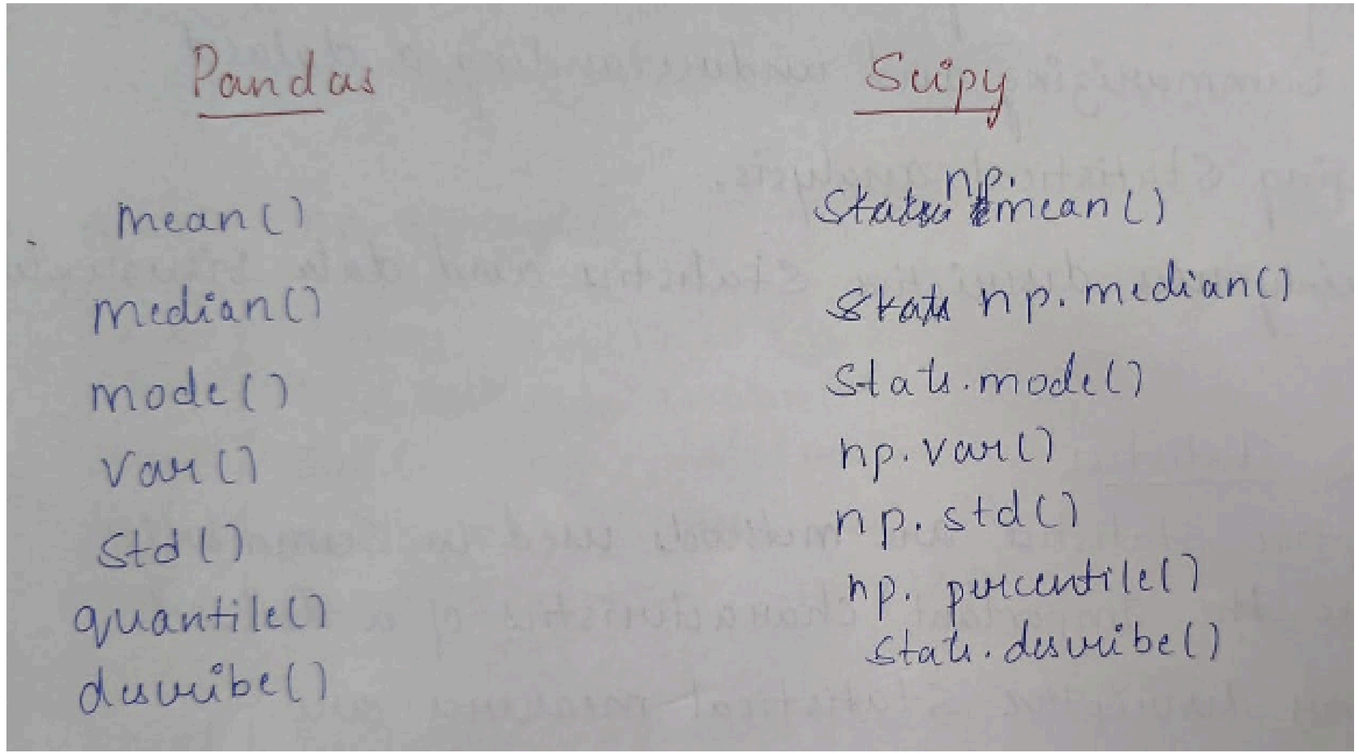
marks = np.array([                ])
print("Mean:", np.mean(marks))
print("Median:" np.median(marks))
print("Mode:", np.mode(marks))
print("Variance:", np.var(marks))
print("Standard Deviation:", np.std(marks))

print("Minimum:", np.min(marks))
print("Maximum:", np.max(marks))

print("25th percentile:", np.percentile(marks,25))
print("50th percentile:", np.percentile(marks,50))
print("75th percentile:", np.percentile(marks,75))

print(stats.describe(marks))
```

Note: Pandas use Quantile(), while Numpy use percentile() because pandas use a decimal value like 0.25, while Numpy use a percentage value like 25.



"Pandas is mainly for handling and analyzing tabular data, while SciPy provides a wide range of scientific and statistical functions that we can use for deeper analysis."

Data distributions: normal, skewed, kurtosis; probability distributions

What is Data Distribution?

"Suppose I ask the marks of 50 students in your class. Will all 50 students get exactly the same marks?"
No.
"So how are their marks spread?"

For example:
35, 42, 45, 48, 50, 52, 53, 55, 57, 58,
60, 61, 62, 63, 64, 65, 67, 70, 72, 85

"These numbers are not randomly placed. Some marks occur around a particular range, while fewer students have very low or very high marks."

That pattern of how the values are **spread** is called a **distribution**.

Distribution means the way data values are spread across different values or ranges.

Why do we care about the distribution?

"If I only tell you that the average mark is 60, do you know everything about the students' marks?"

Now take two examples.
Class A
55, 57, 58, 60, 61, 62, 65
Class B
20, 30, 40, 60, 80, 90, 100

Both could have an average around 60, but the **spread of the data is completely different**.

"Therefore, the average alone doesn't tell us the complete story. We also need to understand how the data is distributed."

This is the reason we study data distributions.

From a distribution, we can understand:

- **Where most values occur**
- **Whether values are spread widely or narrowly**
- **Whether the data is balanced**
- **Whether extreme values exist**
- **Whether the data is approximately normal or skewed**

1. Normal Distribution

Normal distribution is a symmetric, bell-shaped distribution in which most observations are concentrated around the center and fewer observations occur toward the extremes.

"Suppose I measure the heights of 1,000 people. Will most people have extremely low or extremely high height?"
No.
Most people will be somewhere around the average.
For example:

```
150  155  160  165  170  175  180  185  190
 |    |    |    |    |    |    |    |    |
            MOST PEOPLE
                 ↓
                170
```

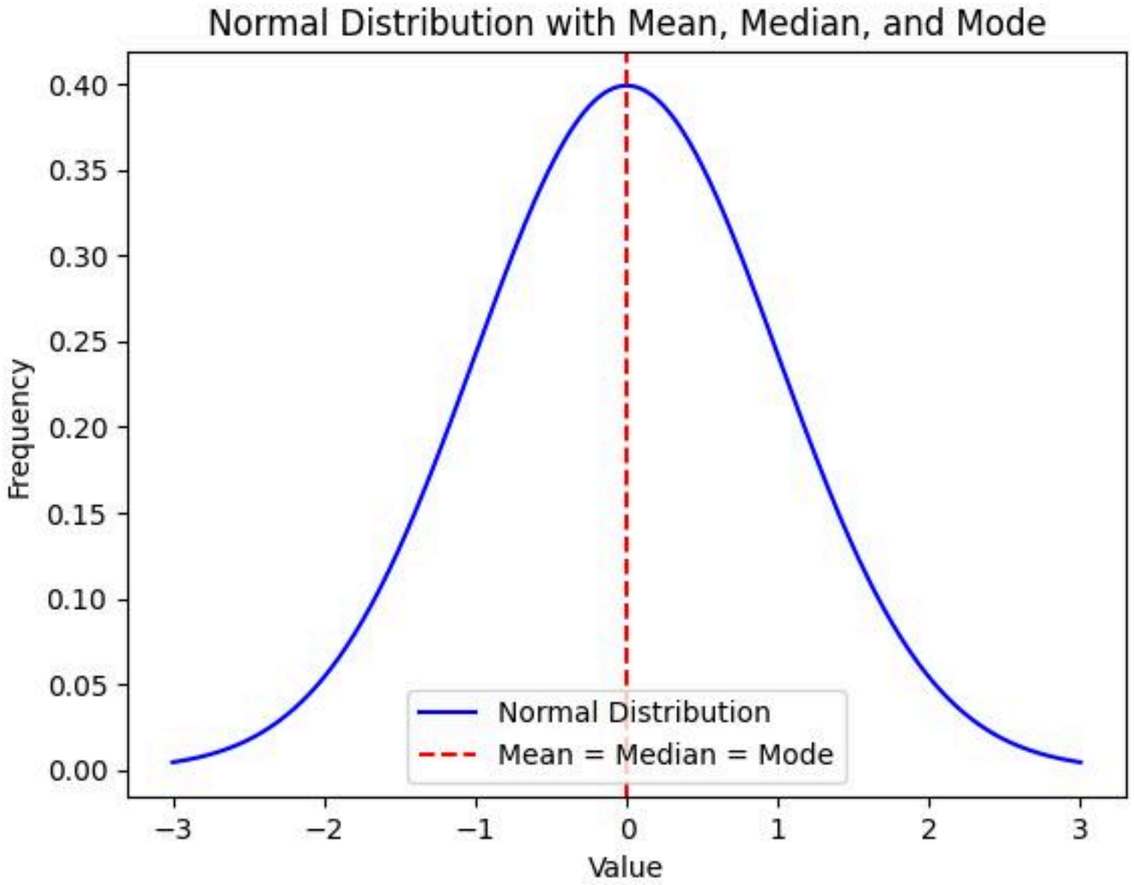
The basic pattern is:

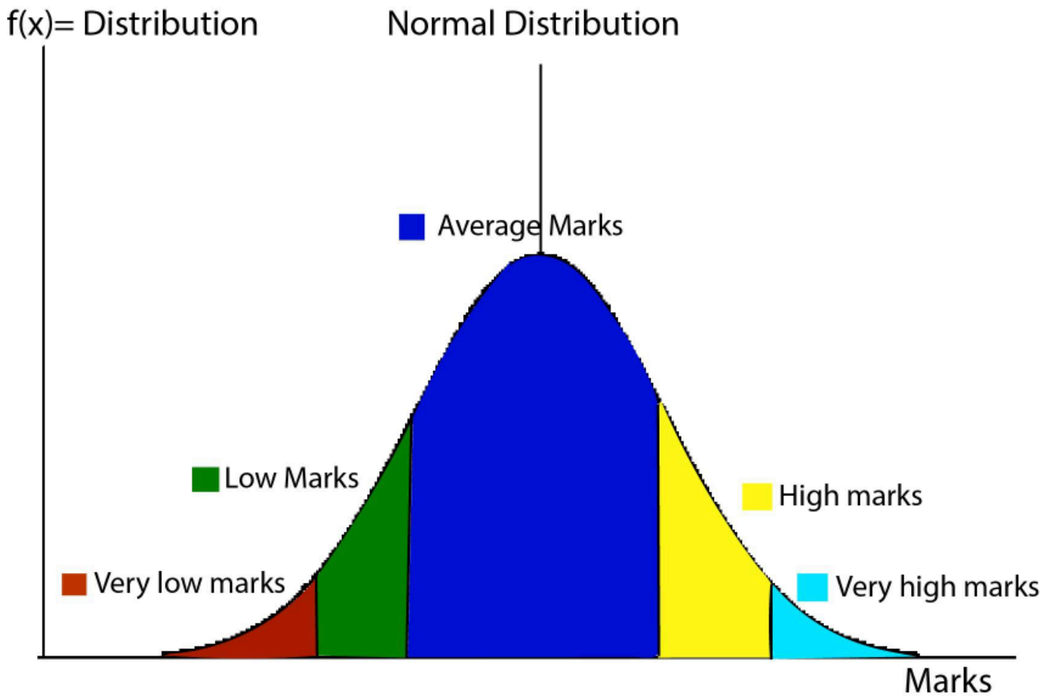
```
Few values        Many values        Few values
   ↓                  ↓                  ↓


__________          /\          __________
          \        /  \        /
           \______/    \______/
```

Most people are around 170 cm.
Very few people are at the extreme ends.

When we plot this, we get approximately a **bell shape**.





"Normal distribution means the data is approximately balanced around the center and has a bell-shaped pattern."

bell shape



"The middle is high because most observations are near the average."
"As we move away from the center, fewer observations occur."
"Therefore, the curve becomes lower on both sides."
That's why we get a **bell shape**.

Mean, Median and Mode in Normal Distribution

In a perfectly normal distribution:

Mean = Median = Mode

Mean

Average value.

Median

Middle value when the data is arranged in order.

Mode

Most frequently occurring value.

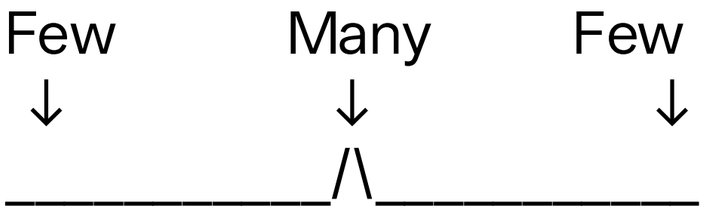
"In a perfectly symmetrical normal distribution, all three meet at the center."

A real-life example

"Imagine we collect the heights of 10,000 people."

We might see something like:
Very short people → Few
Average-height people → Many
Very tall people → Few

Therefore:



"This is the basic idea behind a normal distribution."

2. Skewness

Skewness measures the asymmetry of a data distribution and tells us whether the distribution has a longer tail on the left or right side.

What is it?
Sometimes data is **not balanced**.
One side may have a longer tail than the other.

That imbalance is called **skewness**.

There are three important types:

- **Positive skewness → Right tail**
- **Negative skewness → Left tail**
- **Zero Skewness -> Symmetric**

Positive Skewness

Positive skewness occurs when a distribution has a longer tail toward the right, or toward larger values.

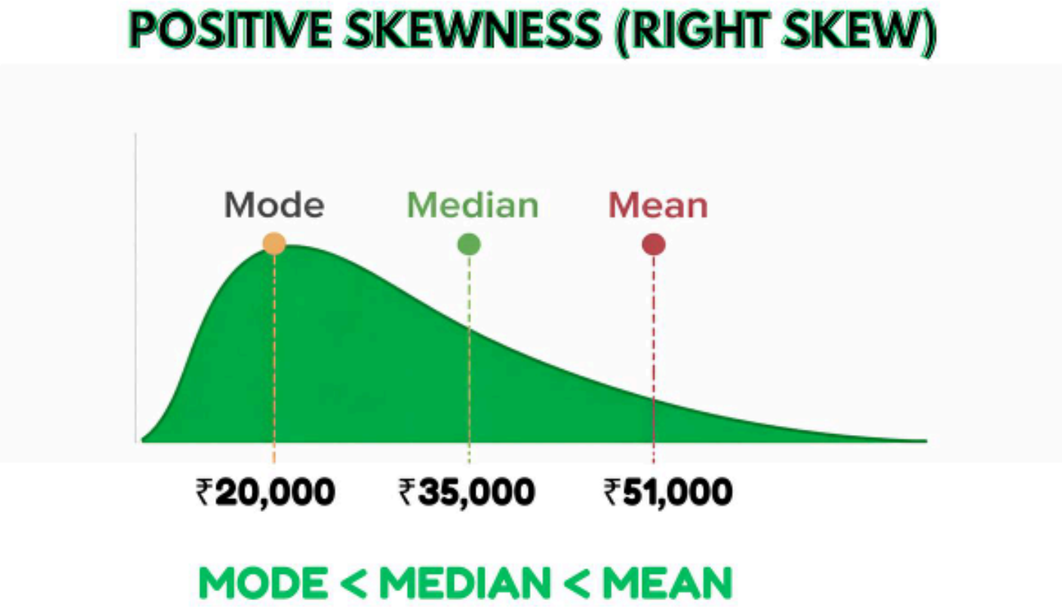
What is it?

Consider salaries:
20K, 22K, 25K, 27K, 30K, 32K, 35K, 40K, 5L

Most employees earn between 20K and 40K.

But one person earns 5 lakh.
That very high value creates a **long right tail**.

How does it work?

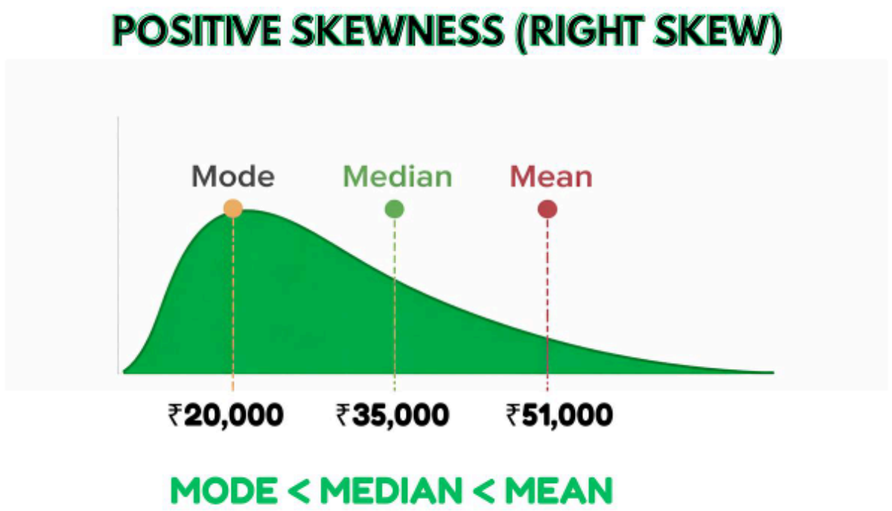


- The green curve represents how the **income values are distributed**.
- Most people are concentrated around the **lower income range**, while a small number of people have **very high incomes**.

For example:
₹ 20,000, ₹ 22,000, ₹ 25,000, ₹ 27,000, ₹ 30,000, ₹ 35,000, ₹ 40,000, ₹ 50,000, ₹ 80,000

Here, most values are relatively low, but a few values are very high.
Those high values create a **long tail toward the right side**.

That's why it is called **right skewness**.



Suppose we have incomes:
₹ 20K, ₹ 22K, ₹ 25K, ₹ 27K, ₹ 30K, ₹ 35K, ₹ 40K, ₹ 50K, ₹ 100K
Most incomes are around ₹ 20K– ₹ 40K.

But there is one very high income:
₹ 100K

That very large value **pulls the average toward the right**.

So, the mean increases.
That's why:
Mode < Median < Mean

In the diagram:
₹ 20,000 < ₹ 35,000 < ₹ 51,000

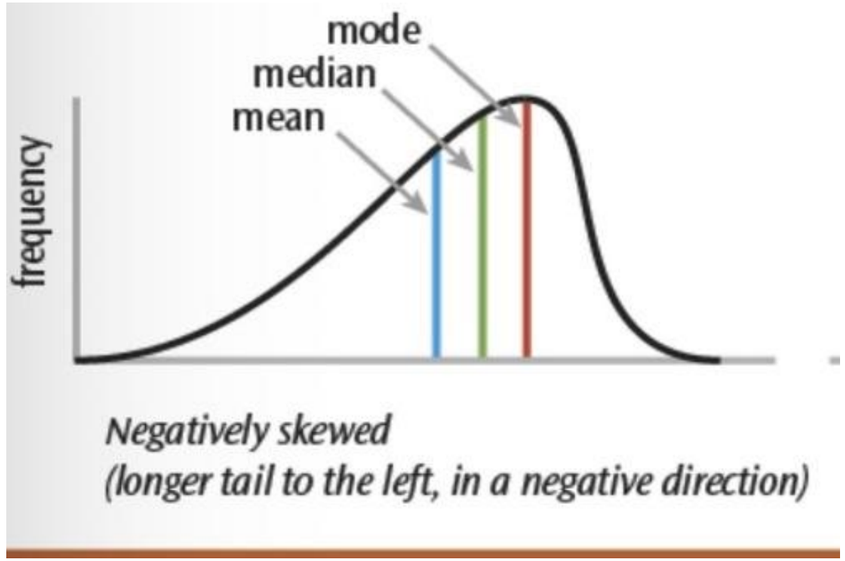
Negative Skewness

Negative skewness occurs when a distribution has a longer tail toward the left, or toward smaller values.

What is it?

Imagine an **easy examination**.
Most students get high marks:
75, 80, 85, 88, 90, 92, 95, 98
But a few students get:
30, 40
The few low marks create a **left tail**.

How does it work?



- The curve represents a distribution of data values.
- Here, **most of the data values are concentrated on the right side**, near the higher values.
- But there are a few **very small values** on the left side.
- These small values create a **long tail toward the left**.

So:

Long tail → Left → Negative skewness

The order is:

Mean < Median < Mode

Imagine students' marks:
20, 70, 75, 78, 80, 82, 85, 88, 90
Most students scored between **70 and 90**.
But one student scored very low: **20**.
That very low value pulls the **average (mean)** toward the left.

So:

- **Mean** gets pulled toward the low value.
- **Median** stays around the middle.
- **Mode** remains near the most common/highe concentration.

Therefore:
Mean < Median < Mode

Zero skewness

Zero skewness means the data is balanced equally on the left and right side of the center.

When **skewness = 0**, the data is **perfectly symmetric**.

Simple example

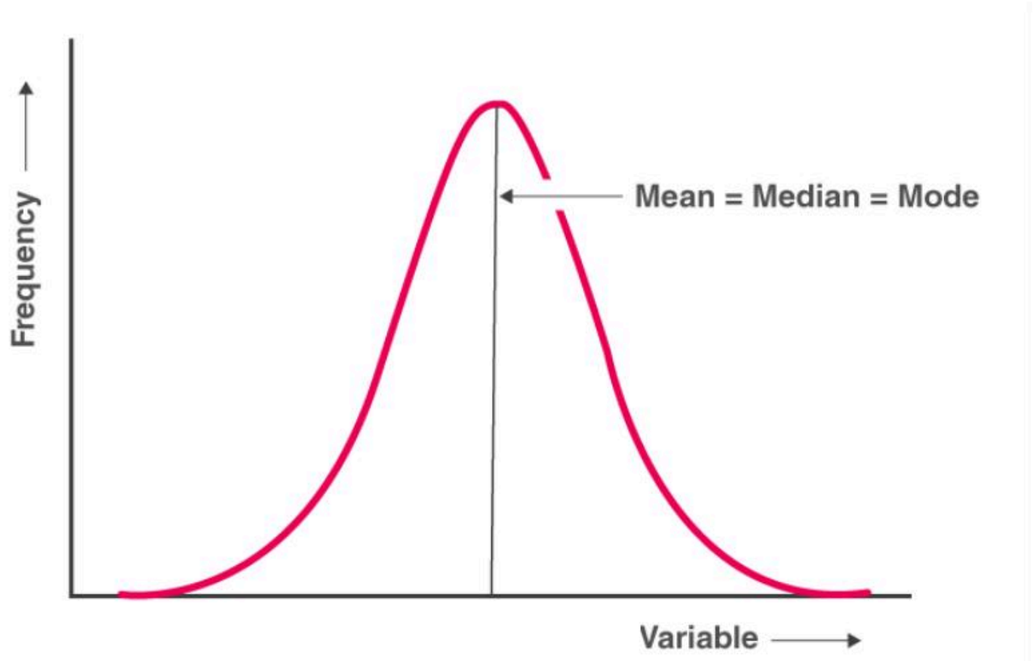
Suppose the marks are:

40, 45, 50, 55, 60

Here:

- Mean = **50**
- Median = **50**
- The values below 50 and above 50 are balanced.
- There is no long tail on either side.

So, **skewness = 0**.



For a perfectly symmetric distribution:

Mean = Median = Mode

Measurement of Skewness

"Now we know what skewness tells us. But how do we actually calculate it?"

There are different methods for measuring skewness. They are:

1.Karl Pearson's coefficient of skewness

2.Bowley's coefficient of skewness

3.Kelly’ s coefficient of skewness

1. Karl Pearson's Coefficient of Skewness

Karl Pearson's coefficient of skewness is a method used to measure the **direction and degree of asymmetry** in a dataset using mean , mode, median and standard deviation.

In simple words:
It tells us whether the data is symmetric, shifted toward the right, or shifted toward the left.

Formula

When **Mode** is available:

$$Sk = \frac{Mean - Mode}{Standard\ Deviation}$$

If **Mode** is not available, we can use:

$$Sk = \frac{3(Mean - Median)}{Standard\ Deviation}$$

Problem 1: Mode is Available

Problem

The marks obtained by 7 students are:

$$10,\ 20,\ 20,\ 20,\ 30,\ 40,\ 50$$

Calculate Karl Pearson's coefficient of skewness.

Since **20 occurs three times**, the mode is clearly available.

Formula

$$Sk = \frac{Mean - Mode}{Standard\ Deviation}$$

Step 1: Calculate Mean

$$Mean = \frac{10 + 20 + 20 + 20 + 30 + 40 + 50}{7}$$

$$= \frac{190}{7}$$

Mean=27.14

$$\text{Mean } (\bar{x}) = \frac{\sum x}{n}$$

Step 2: Find Mode

20 occurs most frequently.

Mode=20

Step 3: Calculate Standard Deviation

We use:

$$SD = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

x	$X - 27.14$	$(X - 27.14)^2$
10	-17.14	293.78
20	-7.14	50.98
20	-7.14	50.98
20	-7.14	50.98
30	2.86	8.18
40	12.86	165.38
50	22.86	522.58

$$\sum (X - \bar{X})^2 = 1142.86$$

Therefore:

$$SD = \sqrt{\frac{1142.86}{7}}$$

$$SD = \sqrt{163.27}$$

SD≈12.78

Step 4: Calculate Skewness

$$Sk = \frac{Mean - Mode}{SD}$$

$$Sk = \frac{27.14 - 20}{12.78}$$

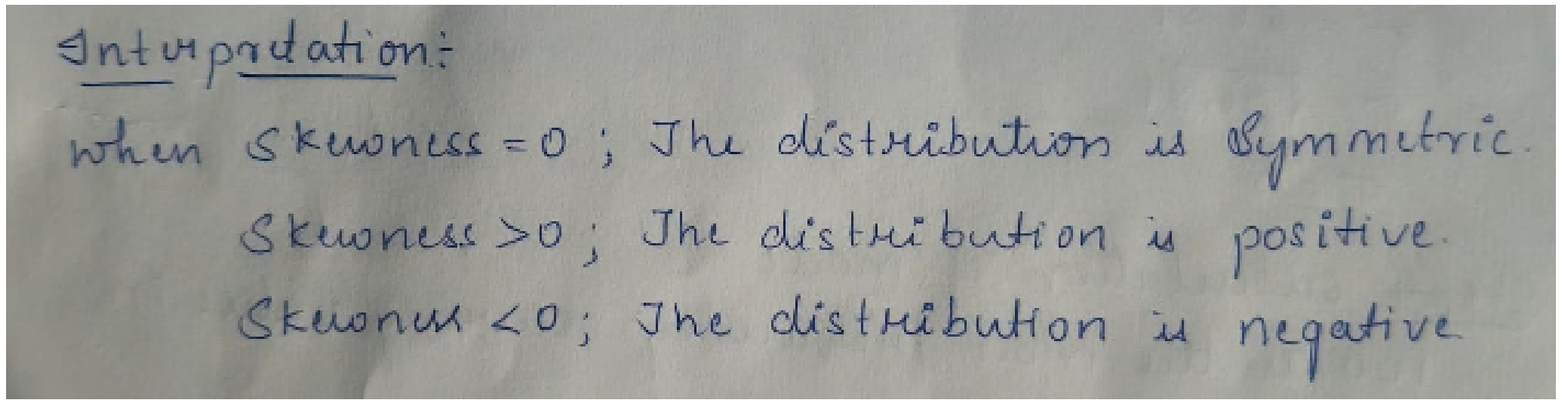
$$Sk = \frac{7.14}{12.78}$$

Sk≈+0.56

Interpretation

Since skewness is **positive**:

The distribution is **positively skewed**, meaning it has a tendency toward the right side.



Problem 2: Mode is Unavailable

Problem

The marks obtained by 7 students are:

10, 20, 30, 40, 50, 60, 100

Calculate Karl Pearson's coefficient of skewness.

Now look at the data:

- 10 occurs once
- 20 occurs once
- 30 occurs once
- 40 occurs once
- 50 occurs once
- 60 occurs once
- 100 occurs once

There is **no value that occurs more frequently than the others**.

Therefore:

There is no mode.

So we **cannot use**:

$$Sk = \frac{Mean - Mode}{SD}$$

Instead, we use the alternative formula:

$$Sk = \frac{3(Mean - Median)}{SD}$$

Step 1: Calculate Mean

$$Mean = \frac{10 + 20 + 30 + 40 + 50 + 60 + 100}{7}$$

$$= \frac{310}{7}$$

$$\boxed{Mean = 44.29}$$

$$\text{Mean } (\bar{x}) = \frac{\sum x}{n}$$

Step 2: Calculate Median

The data is already arranged:

10, 20, 30, 40, 50, 60, 100

There are 7 observations. The middle value is 40

Therefore:

$$\boxed{Median = 40}$$

Step 3: Calculate Standard Deviation

Use:

$$SD = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

Mean = 44.29.

x	$X - 44.29$	$(X - 44.29)^2$
10	-34.29	1175.84
20	-24.29	590.00
30	-14.29	204.08
40	-4.29	18.37
50	5.71	32.65
60	15.71	246.94
100	55.71	3103.88

$$\sum (X - \bar{X})^2 \approx 5371.76$$

Therefore:

$$SD = \sqrt{\frac{5371.76}{7}}$$

$$SD = \sqrt{767.39}$$

SD≈27.70

Step 4: Calculate Skewness

Use:

$$Sk = \frac{3(Mean - Median)}{SD}$$

Substitute:

$$Sk = \frac{3(44.29 - 40)}{27.70}$$

$$= \frac{3(4.29)}{27.70}$$

$$= \frac{12.87}{27.70}$$

$$\boxed{Sk \approx +0.46}$$

Interpretation

Since:

$$Sk = +0.46$$

the distribution is **positively skewed**.

The value **100** is much higher than the other marks, so it stretches the distribution toward the **right**.

2. Bowley's Coefficient of Skewness

Bowley's coefficient of skewness is a method used to measure the **asymmetry of a distribution using quartiles and median**.

In simple words:
Bowley's method checks whether the data is balanced around the median by comparing the lower part and upper part of the data.

It is especially useful when there are **extreme values**, because it uses quartiles rather than mean and standard deviation.

Formula

$$Sk_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Where:
- Q_1 =First quartile
- Q_2 =Median
- Q_3 =Third quartile

Problem

Consider the following marks:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60,\ 80$$

Calculate Bowley's coefficient of skewness.

Step 1: Arrange the data

The data is already arranged in ascending order:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60,\ 80$$

There are:

$$N = 7$$

observations.

Step 2: Find Q_2 –Median

The formula for the position of the median is:

$$\frac{N+1}{2}$$

Substitute:

$$\frac{7+1}{2} = \frac{8}{2} = 4$$

So, the **4th value** is the median.

The 4th value is:

Q2=40

Step 3: Find Q_1

For Q_1 ,take the values **below the median**.

Our original data is:

$$10,\ 20,\ 30,\ \boxed{40},\ 50,\ 60,\ 80$$

The lower half is:

$$10,\ 20,\ 30$$

The middle value of these three values is:

$$\boxed{Q_1 = 20}$$

So:

Q1 =20

Step 4: Find Q_3

Now take the values **above the median**.

Upper half:

$$50,\ 60,\ 80$$

The middle value is:

$$\boxed{Q_3 = 60}$$

So now we have:

$$Q_1 = 20$$
$$Q_2 = 40$$
$$Q3 = 60$$

Step 5: Apply Bowley's formula

Formula:

$$Sk_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Substitute the values:

$$Sk_B = \frac{60 + 20 - 2(40)}{60 - 20}$$

First calculate the numerator:

$$60 + 20 - 80$$
$$= 0$$

Now calculate the denominator:

$$60 - 20 = 40$$

Therefore:

$$Sk_B = \frac{0}{40}$$

SkB=0

Step 6: Interpretation

We obtained:

$$\boxed{Sk_B = 0}$$

Therefore, the distribution is **symmetric** according to Bowley's measure.

In simple words:

The distance from Q_1 to the median is the same as the distance from the median to Q_3.

Let's see:

$$Q_2 - Q_1 = 40 - 20 = 20$$

and

$$Q_3 - Q_2 = 60 - 40 = 20$$

Both are equal.

Therefore, the data is balanced around the median.

Let's Take a Positive Skewness Example

Consider:

$$10,\ 20,\ 30,\ 40,\ 50,\ 70,\ 100$$

Find Bowley's coefficient of skewness.

Step 1: Find Q_2

There are 7 observations.

Median position:

$$\frac{7+1}{2} = 4$$

4th value:

Q2=40

Step 2: Find Q_1

Lower half:

$$10,\ 20,\ 30$$

Middle value:

$$\boxed{Q_1 = 20}$$

Step 3: Find Q_3

Upper half:

$$50,\ 70,\ 100$$

Middle value:

$$\boxed{Q_3 = 70}$$

Therefore:

$$Q_1 = 20, Q_2 = 40, Q_3 = 70$$

Step 4: Apply the formula

$$Sk_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$= \frac{70 + 20 - 2(40)}{70 - 20}$$

$$= \frac{70 + 20 - 80}{50}$$

$$= \frac{10}{50}$$

SkB=0.2

Step 5: Interpret

Since:

$$Sk_B = +0.2$$

the distribution is **positively skewed**.

Why?

Look at the distances:

Left side of median

$$Q_2 - Q_1 = 40 - 20 = 20$$

Right side of median

$$Q_3 - Q_2 = 70 - 40 = 30$$

The upper side is more spread out.

Therefore, the distribution has more stretch toward the **right**.

Bowley's coefficient	Interpretation
$Sk_B = 0$	Symmetric
$Sk_B > 0$	Positively / right skewed
$Sk_B < 0$	Negatively / left skewed

3. Kelly's Coefficient of Skewness

Kelly's coefficient of skewness is a method used to measure the **direction and degree of skewness using percentiles**.

In simple words:
Kelly's method checks whether the distribution is balanced by comparing the lower and upper parts of the data using percentiles.

Unlike:

- **Karl Pearson** → uses Mean, Mode/Median and SD
- **Bowley** → uses Q_1 , Median and Q_3
- **Kelly** → uses **P_{10} , P_{50} and P_{90}**

Formula

Kelly's coefficient of skewness is:

$$Sk_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

Where:

- P_{10} = 10th percentile
- P_{50} = 50th percentile (Median)
- P_{90} = 90th percentile

Why P_{10}, P_{50} and P_{90}?

"Kelly's method looks at a wider portion of the distribution than Bowley's method."

Bowley looks at:

$$Q_1 = P_{25}, Q_2 = P_{50}, Q_3 = P_{75}$$

Kelly looks at:

$$P_{10}, P_{50}, P_{90}$$

So, Kelly considers **90% of the distribution**, rather than focusing only on the middle 50%.

Example Problem

Consider the following dataset:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60,\ 70,\ 80,\ 90$$

Calculate **Kelly's coefficient of skewness**.

Step 1: Arrange the data

The data is already arranged in ascending order:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60,\ 70,\ 80,\ 90$$

Number of observations:

N=9

Step 2: Find P_{50}

P_{50} is the **50th percentile**, which is the same as the median.

$$Position(P_k) = \frac{k(N+1)}{100}$$

Position of P_{50}:

$$\frac{50(N+1)}{100}$$

Substitute $N = 9$:

$$P_{50} = \frac{50(9+1)}{100}$$

$$= \frac{50(10)}{100}$$

$$= 5$$

So, we take the **5th value**.

The 5th value is:

P50=50

Step 3: Find P_{10}

Formula for percentile position:

$$Position(P_k) = \frac{k(N+1)}{100}$$

For P_{10}:

$$P_{10} = \frac{10(9+1)}{100}$$
$$= \frac{100}{100}$$
$$= 1$$

Therefore, the 1st value is:

$$\boxed{P_{10} = 10}$$

Step 4: Find P_{90}

Using the same formula:

$$P_{90} = \frac{90(N+1)}{100}$$
$$= \frac{90(9+1)}{100}$$
$$= \frac{900}{100}$$
$$= 9$$

Therefore, the 9th value is:

P90=90

Step 5: Now we have all required values

We have:

$$P_{10} = 10$$
$$P_{50} = 50$$
$$P_{90} = 90$$

Now apply Kelly's formula:

$$Sk_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

Substitute:

$$Sk_K = \frac{90 + 10 - 2(50)}{90 - 10}$$

Step 6: Calculate the numerator

$$90 + 10 - 2(50)$$
$$= 100 - 100$$
$$= 0$$

Step 7: Calculate the denominator

$$90 - 10 = 80$$

Step 8: Calculate Kelly's coefficient

$$Sk_K = \frac{0}{80}$$

Therefore:

SkK=0

Step 9: Interpretation

Since:

$$Sk_K = 0$$

the distribution is **symmetric**.

Why?

Look at the distances from the median:

From P_{50} to P_{10}:

$$P_{50} - P_{10}$$
$$= 50 - 10$$
$$= 40$$

From P_{50} to P_{90}:

$$P_{90} - P_{50}$$
$$= 90 - 50$$
$$= 40$$

Both distances are equal.

Therefore, the distribution is balanced around the median.

Positive Skewness Example

Consider:

$$10,\ 20,\ 30,\ 40,\ 50,\ 60,\ 70,\ 80,\ 120$$

Here, the high value **120** stretches the data toward the right.

For $N = 9$:

$$P_{10}$$

$$P_{10} = \frac{10(10)}{100} = 1$$

So:

$$\boxed{P_{10} = 10}$$

$$P_{50}$$

$$P_{50} = \frac{50(10)}{100} = 5$$

So:

$$\boxed{P_{50} = 50}$$

$$P_{90}$$

$$P_{90} = \frac{90(10)}{100} = 9$$

So:

P90=120

Apply the formula

$$Sk_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

Substitute:

$$Sk_K = \frac{120 + 10 - 2(50)}{120 - 10}$$

$$= \frac{130 - 100}{110}$$

$$= \frac{30}{110}$$

SkK≈+0.27

Interpretation

Since:

$$Sk_K = +0.27$$

the distribution is **positively skewed**.

The reason is that P_{90} is much farther from the median than P_{10} is.

$$P_{90} - P_{50} = 120 - 50 = 70$$

while:

$$P_{50} - P_{10} = 50 - 10 = 40$$

So the upper side is more stretched.

Therefore:

The distribution has a longer right-side tendency.

Comparison of Three Measures

Measure	Values used	Main idea
Karl Pearson	Mean, Mode/Median, SD	Compares central measures
Bowley	Q_1, Median, Q_3	Uses quartiles
Kelly	P_{10}, P_{50}, P_{90}	Uses percentiles

3. Kurtosis

Kurtosis describes the heaviness of the tails of a distribution and is related to how prominent extreme observations are.

What is it?

Skewness

Which direction does the distribution's tail extend?

Kurtosis

How heavy are the tails and how prominent are extreme observations?

Example of Kurtosis

Suppose we have two datasets.

Dataset A

45, 48, 50, 52, 54, 55, 57

There are no very extreme values.

Dataset B

20, 48, 50, 52, 54, 55, 90

Here we have extreme values:

20	90
↓	↓
Extreme	Extreme

Kurtosis helps us describe the **tail behavior and extreme observations**.

Types of Kurtosis

A. Mesokurtic

A distribution with tail behavior approximately like the normal distribution is called mesokurtic.

B. Leptokurtic

A leptokurtic distribution has heavier tails, meaning extreme observations are more prominent.

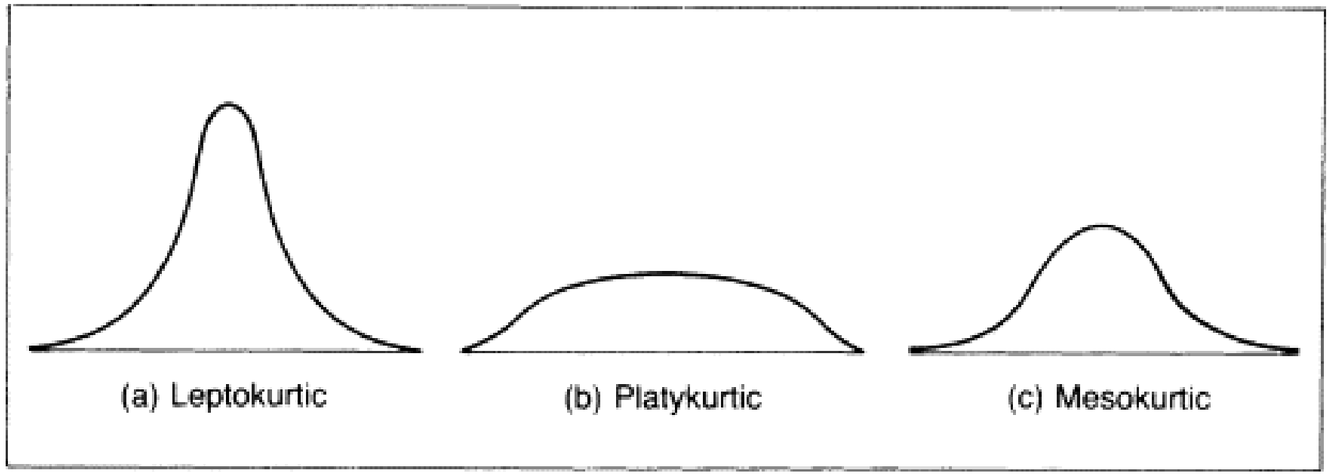
Leptokurtic = Heavy tails

C. Platykurtic

A platykurtic distribution has lighter tails, meaning extreme observations are less prominent.

Simple memory:

Platykurtic = Light tails



Type	Meaning	Excess kurtosis
Mesokurtic	Similar kurtosis to a normal distribution	0
Leptokurtic	Heavier tails than a normal distribution	Positive
Platykurtic	Lighter tails than a normal distribution	Negative

4. Probability Distribution

Probability

Probability is a measure of how likely an event is to occur.

What is it?

Suppose I toss a coin.
Can I guarantee that it will be Head?
No.
There are two possible outcomes:
Head
Tail

For a fair coin:
P(Head) = 0.5
P(Tail) = 0.5

How does it work?

Probability ranges from:
0 → Impossible
1 → Certain

For example:
Probability = 0

→ Event cannot happen

Probability = 0.5

→ Equal chance

Probability = 1

→ Event is certain

Probability Distribution

A probability distribution describes all possible values or outcomes of a random variable and assigns a probability to each outcome.

What is it?

Let's roll a fair dice.
Possible outcomes:
1, 2, 3, 4, 5, 6
Each outcome has:
Probability = 1/6
So:

Dice outcome	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

This table is a **probability distribution**.

How does it work?

Think of it as:
Random experiment
↓
Possible outcomes
↓
Probability of each outcome
↓
Probability distribution

Probability Distribution

Definition: A probability distribution describes all possible outcomes of a random variable and assigns a probability to each outcome.

Example: Roll a fair dice

The possible outcomes are 1, 2, 3, 4, 5 and 6.



Probability Distribution	
Outcome	**Probability**
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

What it tells us:

- The possible outcomes (1 to 6).
- The probability of each outcome (1/6).

This table is a probability distribution.

How it works:

1. **Identify the random experiment** (e.g., roll a dice) →
2. **List all possible outcomes** (1, 2, 3, 4, 5, 6) →
3. **Assign probabilities to each outcome** (1/6 each) →
4. **Get the probability distribution** (the complete table)

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Correlation and covariance; Introduction to hypothesis testing basics (t-test, chi-square)

Case study: Student Performance Analysis

Problem Statement

A college wants to understand the performance of its students.
The college has information such as **Study Hours, Attendance, Previous Marks, and Final Marks**.

The teacher wants to answer:
1.Are Study Hours and Marks related?
2.Is there a relationship between Attendance and Marks?
3.Did a training program improve students' marks?
4.Is there any relationship between Gender and Course Preference?

From this one case, we will be going to learn:
Correlation → Covariance → Hypothesis Testing → t-test → Chi-square

Correlation

"If a student studies more hours, do you think their marks will also increase?"
yes.
"We can use correlation to measure this relationship. "

Simple definition:
Correlation tells us whether two numerical variables are related and how strongly they are related. (OR)
Correlation tells us both the direction AND the strength of the relationship.

Example:

Study Hours	Marks
2	45
4	55
6	65
8	80
10	90

Here, when Study Hours increase, Marks also increase.
So, we have a **positive correlation**.

Correlation answers: "When one variable changes, does the other variable also change?"

For example, consider:

- Study Hours
- Exam Marks

If students who study more generally get higher marks, then **Study Hours and Marks have positive correlation**.

Correlation value

Correlation usually ranges from **-1 to +1**.

- **+1** → Perfect positive relationship
- **0** → No linear relationship
- **-1** → Perfect negative relationship

For example:

Correlation = 0.85
This means there is a **strong positive relationship**.
Correlation = -0.80
This means there is a **strong negative relationship**.
Correlation = 0.05
This means there is **almost no linear relationship**.

Three possibilities

Correlation	Meaning	Example
Positive	Both increase or decrease together	Study Hours ↑ → Marks ↑
Negative	One increases while the other decreases	Price ↑ → Demand ↓
Zero/No correlation	No clear relationship	Shoe size and exam marks

```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

df = pd.DataFrame({
    "StudyHours": [2, 4, 6, 8, 10],
    "Marks": [45, 55, 65, 80, 90]
})

# Calculate correlation
correlation = df["StudyHours"].corr(df["Marks"])

print("Correlation:", correlation)

# Scatter plot
plt.scatter(df["StudyHours"], df["Marks"])

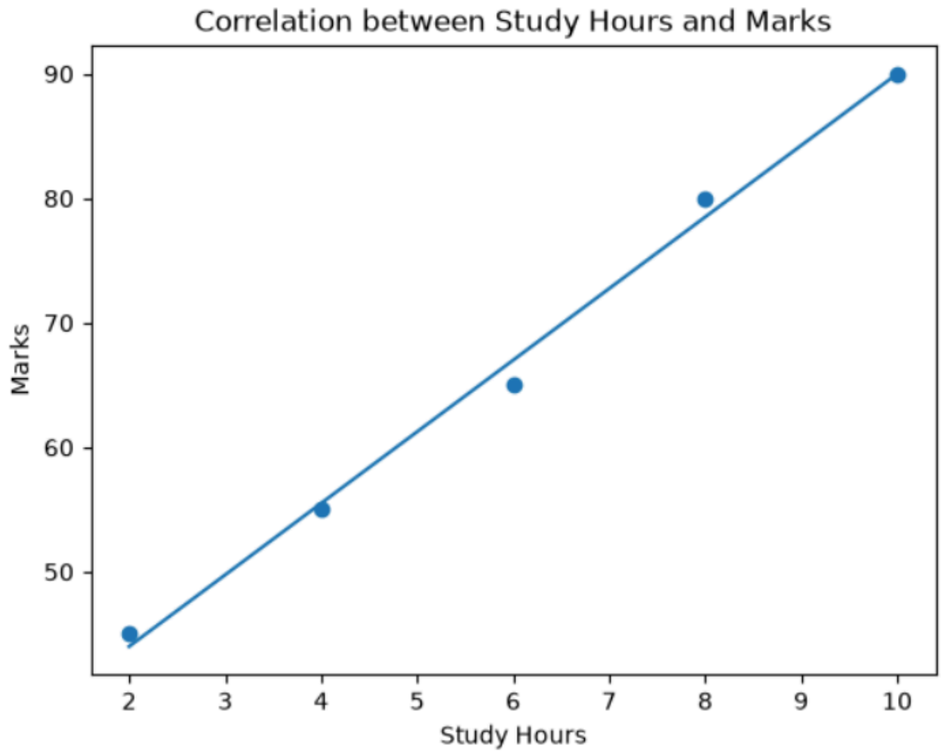
# Trend line
z = np.polyfit(df["StudyHours"], df["Marks"], 1)
p = np.poly1d(z)

plt.plot(df["StudyHours"], p(df["StudyHours"]))

plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Correlation between Study Hours and Marks")

plt.show()
```

```
Correlation: 0.9971764649527378
```



If the result is close to **+1**, we say:
"There is a strong positive relationship between Study Hours and Marks."

```
z = np.polyfit(df["StudyHours"], df["Marks"], 1)
```

This line calculates the mathematical equation of the **best-fit straight line**.

np.polyfit()
means:
"Find the polynomial line that best fits the given data."

We provide three things:
np.polyfit(X, Y, degree)

Here:
np.polyfit(
df["StudyHours"],
df["Marks"],
1
)

First input
df["StudyHours"]
This is the X variable.
X = Study Hours

Second input
df["Marks"]
This is the Y variable.
Y = Marks

Third input
1
The value 1 means we want a **first-degree polynomial**

A **first-degree polynomial** is simply a **straight line**:
y = mx + c
Where:
- m = slope of the line
- c = intercept

The result is stored in:
z

"np.polyfit() calculates the slope and intercept required to create the best-fit straight line."

```
p = np.poly1d(z)
```

After **polyfit()** calculates the values, we have:
z
But we want to use those values easily as a mathematical function.
Therefore:
np.poly1d(z)
creates a polynomial function.
The function is stored in:
p
Conceptually, it becomes something like:
p(x) = mx + c

Now we can give any StudyHours value to **p()** and it calculates the corresponding value on the trend line.
For example:
p(2)
calculates the trend-line value for 2 study hours.
Similarly:
p(6)
calculates the trend-line value for 6 study hours.

Covariance

"We know that Study Hours and Marks are moving together. Can we measure whether they move in the same or opposite direction?"

Simple definition
Covariance tells us whether two variables move together or move in opposite directions.

For our example:
Study Hours ↑
↓
Marks ↑

Therefore, covariance is **positive**.

Positive covariance means that when one variable increases, the other variable also tends to increase.

Since the covariance is **positive**, Study Hours and Marks tend to move in the **same direction**.

```python
import pandas as pd

# Create the dataset
df = pd.DataFrame({
    "StudyHours": [2, 4, 6, 8, 10],
    "Marks": [45, 55, 65, 80, 90]
})

# Calculate covariance
covariance = df["StudyHours"].cov(df["Marks"])

# Display the result
print("Covariance:", covariance)
```

```
Covariance: 57.5
```

Covariance tells us the direction of the relationship between two variables.

What does +57.5 mean?

Covariance tells us the **direction in which two variables move together**.

- **Positive covariance** → both variables tend to increase together.
- **Negative covariance** → one tends to increase while the other decreases.
- **Covariance close to 0** → little linear co-movement.

So:

Covariance = +57.5

means:

Study Hours and Marks tend to move in the same direction.
When Study Hours increase, Marks also tend to increase.

Three possibilities

Positive covariance

X ↑ → Y ↑

Both variables tend to increase together.

Negative covariance

X ↑ → Y ↓

One increases while the other decreases.

Near-zero covariance

There is no clear linear movement together.

```python
import pandas as pd
import matplotlib.pyplot as plt

# Create the dataset
df = pd.DataFrame({
    "StudyHours": [2, 4, 6, 8, 10],
    "Marks": [45, 55, 65, 80, 90]
})

# Calculate covariance
covariance = df["StudyHours"].cov(df["Marks"])

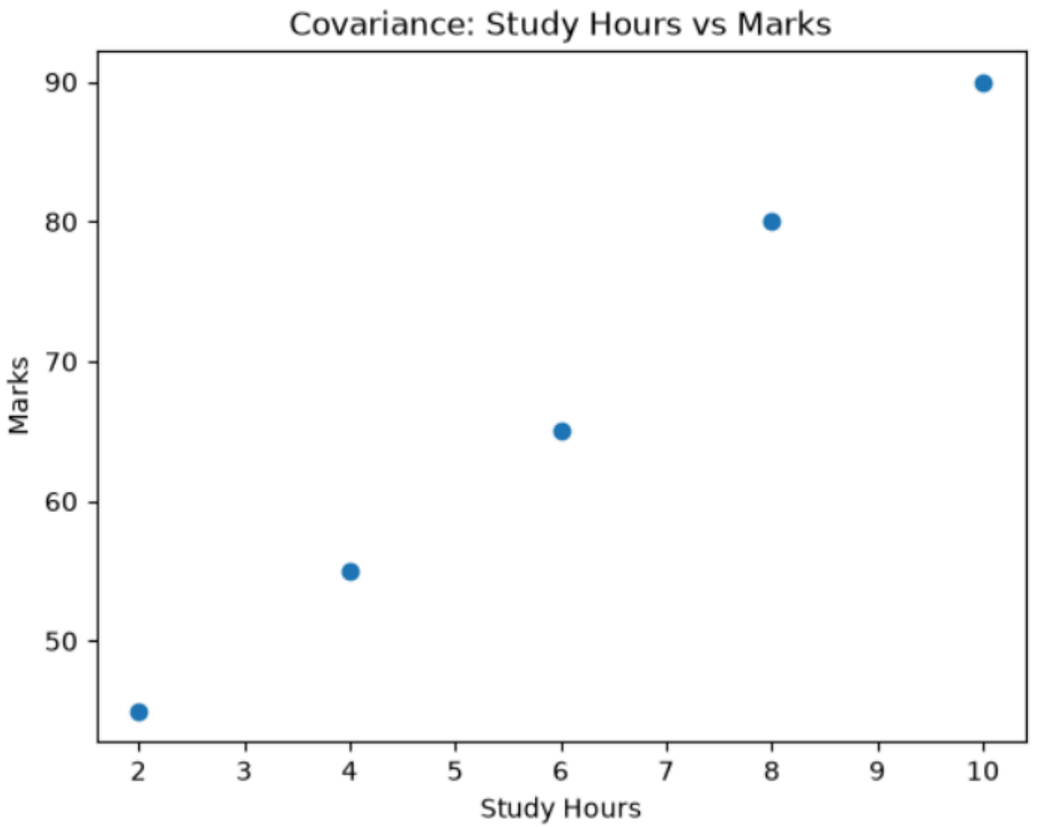
print("Covariance:", covariance)

# Create scatter plot
plt.scatter(df["StudyHours"], df["Marks"])

# Add labels
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Covariance: Study Hours vs Marks")

# Show the graph
plt.show()
```

```
Covariance: 57.5
```



NOTE: Covariance tells us whether two variables move together or in opposite directions, whereas correlation tells us the direction and how strongly they are related.

What is Hypothesis Testing?

"Correlation and covariance helped us understand the relationship between two variables. But sometimes we have a specific claim and want to check whether the data provides enough evidence to support or challenge that claim." (Sometimes we already have a guess or an idea about how things work, and we want to look at the facts to see if our guess is correct.)

Example
Suppose the college says:
"The average marks of our students are 70."
We have some student data.
Our question is:
"Is the college's claim of an average of 70 supported by the sample data?"
We cannot simply say **yes** or **no** by looking at the data.
We need a statistical procedure.
That procedure is called: **Hypothesis Testing**

Simple definition
Hypothesis testing is a statistical method used to test a claim using sample data.

Hypothesis Testing Flow

```
We have a claim
      ↓
We collect sample data
      ↓
We perform a statistical test
      ↓
We get a p-value
      ↓
We make a decision about the claim
```

“Which statistical test should we perform?”

Different Statistical Tests are Used:

The statistical test we choose depends on the type of data and the question we want to answer.

Here, we are going to study two important tests:

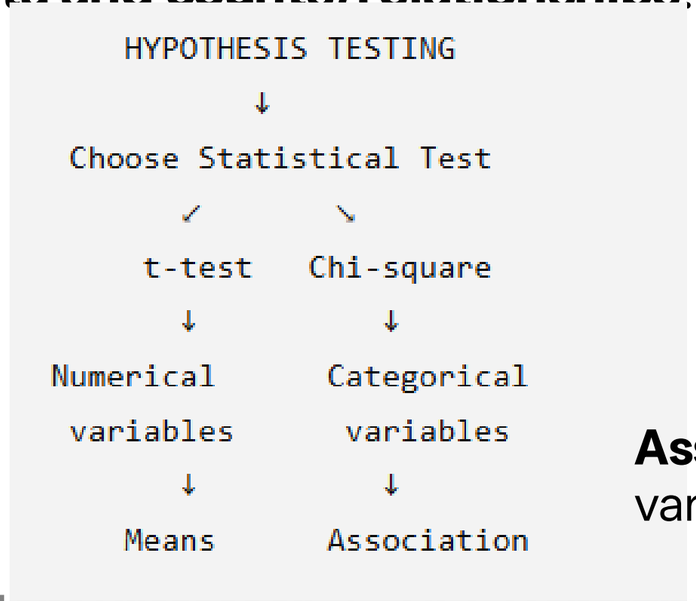
1. t-test

Used mainly with **numerical data and means**.

2. Chi-square test

Used mainly with **categorical data and counts/relationships**.

So, your flow becomes:



Association means, Whether two categorical variables are related to each other or not.

What is a Hypothesis?

A **hypothesis** is a statement about a population that we want to test using data.

In our case:

Claim: Average marks = 70

We create two statements.

- **Null Hypothesis — H_0**

The null hypothesis represents the **default assumption**.

$$H_0: \mu = 70$$

Meaning:

The population average marks are 70.

- **Alternative Hypothesis — H_1**

The alternative hypothesis represents the possibility that the claim is not true.

$$H_1: \mu \neq 70$$

Meaning:

The population average marks are different from 70.

NOTE: H_0 → Default assumption / no significant difference

H_1 → Alternative possibility / difference exists

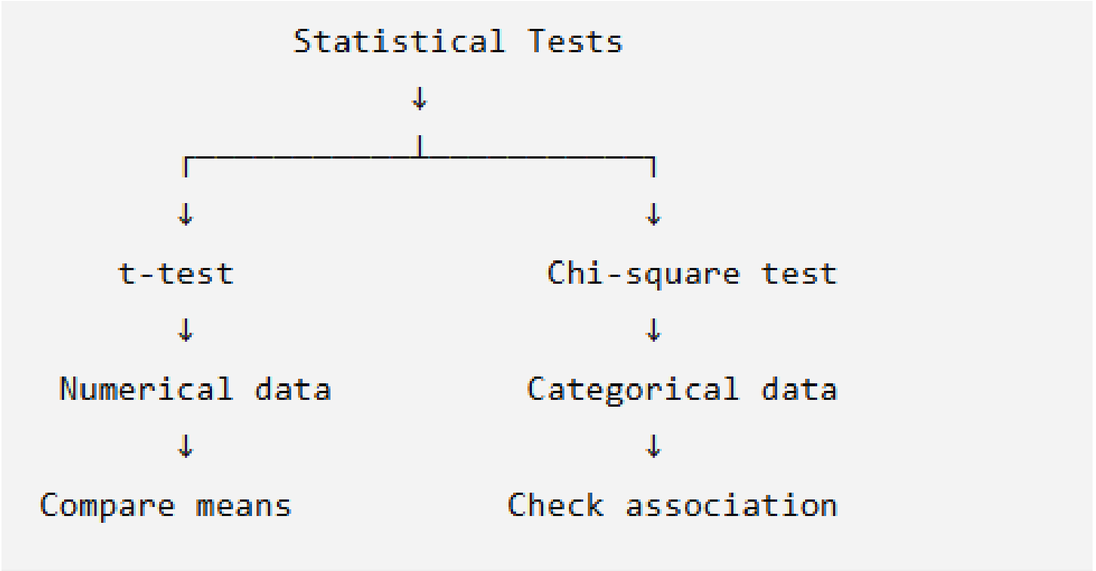
The purpose of the statistical test is to determine whether there is enough evidence to reject H_0 .

How do we actually test H_0 and H_1 ?

Answer:

We use a statistical test.

There are different statistical tests for different situations.



What is a t-test?

A t-test is a statistical test used to determine whether a difference between means is statistically significant.

In simple words:

t-test → used when we are dealing with numerical data and comparing means.

But which t-test should we use?

The type of t-test depends on **what we are comparing**.

Why are there different types of t-tests?

We may have different situations:

Situation 1

We have **one group** and compare its mean with a known/claimed value.

Example:

Are students' average marks different from **70**?

→ **One-Sample t-test**

Situation 2

We have **two different groups**.

Example:

Are Class A and Class B average marks different?

→ **Independent t-test**

Situation 3

We measure the **same students twice**.

Example:

Did training change students' marks?

→ **Paired t-test**

One-Sample t-test: Problem

1. One-Sample t-test

Problem

A teacher claims:

"The average mark of students is 70."

Marks of 5 students:

65, 68, 72, 75, 70

Question

Is the average mark significantly different from 70?

Identify the situation

We have:

- One group of students
- Numerical marks
- A claimed average of 70

Therefore:

One-Sample t-test

Teacher's claim:

$$\mu = 70$$

Sample tells us:

$$\bar{x} = 70$$

Now we ask:

Is the difference between the claimed average and the sample result significant?

Here the difference is:

$$70 - 70 = 0$$

So there is **no difference in the sample average**.

Hypotheses

Step 1: Formulate Hypotheses

H_0 : $\mu = 70$ Null Hypothesis — H_0

Average mark is 70.

H_1 : $\mu \neq 70$ Alternative Hypothesis — H_1

Average mark is different from 70.

We will calculate a **t-value** and **p-value** to decide whether there is enough evidence against H_0 .

Significance level

$\alpha = 0.05$

Step 2: Calculate Mean

$$\bar{x} = \frac{65 + 68 + 72 + 75 + 70}{5} = 70$$

What does this mean?

Our **sample average is 70**.

This is important because the teacher's claimed value is also:

$$\mu_0 = 70$$

So already we can see:

$$\bar{x} - \mu_0 = 70 - 70 = 0$$

But we **cannot stop here**. We still perform the t-test.

Step 3: Calculate Deviation from the Mean

We need the deviation of each mark from the sample mean.

Formula:

$$x - \bar{x}$$

Since:

$$\bar{x} = 70$$

calculate:

Because if we simply add:

$$-5 + (-2) + 2 + 5 + 0 = 0$$

The positive and negative deviations cancel each other.

Mark x	$x - \bar{x}$	$(x - \bar{x})^2$
65	65 − 70 = **−5**	**25**
68	68 − 70 = **−2**	**4**
72	72 − 70 = **2**	**4**
75	75 − 70 = **5**	**25**
70	70 − 70 = **0**	**0**
Total	**0**	**58**

What does deviation mean?

For example:

65:

$$65 - 70 = -5$$

It means 65 is **5 marks below the mean**.

75:

$$75 - 70 = +5$$

It means 75 is **5 marks above the mean**.

Output

We calculate the deviations because we need to know **how spread out the marks are around the mean**.

$$\sum (x - \bar{x})^2 = 58$$

Step 4: Calculate Sample Variance

Formula:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

We already calculated:

$$\sum(x - \bar{x})^2 = 58$$

and:

$$n = 5$$

Therefore:

$$s^2 = \frac{58}{5 - 1}$$

$$s^2 = \frac{58}{4}$$

Output

$$\boxed{s^2 = 14.5}$$

Interpretation

The sample variance is **14.5**.

Step 5: Calculate Standard Deviation

Standard deviation is the square root of variance.

Formula:

$$s = \sqrt{s^2}$$

Substitute:

$$s = \sqrt{14.5}$$

Output

$$\boxed{s \approx 3.81}$$

Interpretation

The marks have a standard deviation of approximately **3.81 marks** around the mean.

Step 6: Calculate the t-Statistic

For a one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Where:

- $\bar{x} = 70$ → sample mean
- $\mu_0 = 70$ → claimed mean
- $s = 3.81$ → sample standard deviation
- $n = 5$ → sample size

Substitute:

$$t = \frac{70 - 70}{3.81/\sqrt{5}}$$

First calculate the numerator:

$$70 - 70 = 0$$

Calculate the denominator:

$$\frac{3.81}{\sqrt{5}} \approx \frac{3.81}{2.236}$$
$$\approx 1.70$$

Therefore:

$$t = \frac{0}{1.70}$$

Output

t=0

The t-value tells us:

How far is our sample mean from the claimed mean, after considering the variability in the sample?

Here:

Sample mean:

$$70$$

Claimed mean:

$$70$$

Difference:

$$70 - 70 = 0$$

Therefore:

$$t = 0$$

In simple words:

Our sample average is exactly equal to the claimed average.

Step 7: Calculate Degrees of Freedom

For a one-sample t-test:

$$df = n - 1$$

Here:

$$n = 5$$

Therefore:

$$df = 5 - 1$$

Output

$$\boxed{df = 4}$$

Interpretation

We have **4 degrees of freedom**.

t-test table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

A **t-distribution table** is used to find the **critical t-value** based on:

- Degrees of freedom
- Significance level α
- One-tailed or two-tailed test

For your example:

- $df = 4$
- $\alpha = 0.05$
- $H_1: \mu \neq 70$

Because H_1 uses $\neq$,
this is a **two-tailed test**.

So, we are asking:
Is the average mark **different from 70?**
It could be:
Greater than 70 → 75, 78, 80, etc.
Less than 70 → 65, 62, 60, etc.
We don't specify a direction.
That's why it is called a **two-tailed test**.

Step 8: Find the critical T value

We have df = 4 and $\alpha = 0.05$

$$\boxed{t_{critical} = 2.776}$$

How 2.776 is obtained

For a two-tailed test with $\alpha = 0.05$:

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

So we need a t-value with **0.025 area in each tail**.

The t-table gives:

$$\boxed{t_{0.025,4} = 2.776}$$

Therefore:

$$\boxed{t_{critical} = \pm 2.776}$$

The rejection regions are:

$$t < -2.776$$

or

$$t > 2.776$$

So the middle region is:

$-2.776 \leq t \leq 2.776$

Step 9: compare t with Critical t value

You calculated:

$$t_{calculated} = 0$$

From the table:

$$t_{critical} = \pm 2.776$$

For a two-tailed test, the rejection regions are:

$$t < -2.776$$

or

$$t > 2.776$$

Your calculated value:

$|t|=0$

$|0|<2.776$

does not fall in either rejection region.

Therefore:

Fail to reject H_0

There is not enough statistical evidence to say that the population mean is different from 70.

Step 10: Find the p-value

We have:

$$t = 0$$
$$df = 4$$

Because this is a **two-tailed test**:

$$H_1: \mu \neq 70$$

the p-value is:

p=1.00

Step 1 — We already calculated the t-value

For the one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

We obtained:

$$t = 0$$

and

$$df = 4$$

Our test is two-tailed because:

$$H_1 : \mu \neq 70$$

What does p-value mean?

The p-value asks:

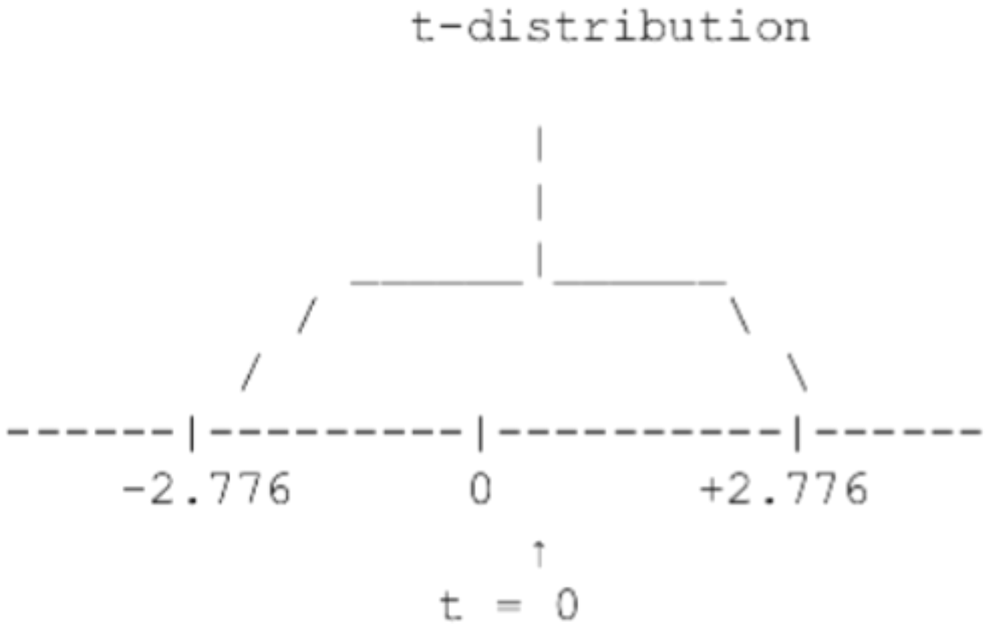
If H_0 is true, how likely is it to get a test statistic as extreme as our observed t-value?

Our observed value is:

$$t = 0$$

The t-distribution is symmetric around 0.

Think of it like this:



Our calculated t-value is **exactly at the center**.

Look at one side first

For a t-distribution:

$$P(T > 0) = 0.5$$

Why?

Because the distribution is perfectly symmetric around zero.

So:

$$P(T > 0) = 0.5$$

and:

$$P(T < 0) = 0.5$$

Therefore:

```
Left side          Right side
   0.5                 0.5
    ↓                   ↓
<--------- 0 ---------->
```

Step 5 — Add both tails

Since this is a **two-tailed test**:

$$p = P(T < -|t|) + P(T > |t|)$$

Our:

$$|t| = |0| = 0$$

Therefore:

$$p = P(T < 0) + P(T > 0)$$

We know:

$$P(T < 0) = 0.5$$

and:

$$P(T > 0) = 0.5$$

Therefore:

$$p = 0.5 + 0.5$$

p=1.00

Why exactly is p-value 1.00?

Our null hypothesis says:

$$H_0: \mu = 70$$

Our sample mean is:

$$\bar{x} = 70$$

So the difference is:

$$\bar{x} - \mu_0 = 70 - 70 = 0$$

Therefore:

$$t = 0$$

A t-value of **0 is exactly at the centre of the t-distribution**.

So the observed result is **not unusual at all** if H_0 is true.

For a two-tailed test, the area corresponding to values at least as extreme as $t = 0$ covers the **entire distribution**:

$$\boxed{p = 1.00}$$

Step 11: Decision
p-value with α (significance level)
For example:
α = 0.05
p-value = 1.00

Since:
1.00>0.05
→ **Reject H_0**

If:
p-value > 0.05
→ **Fail to reject H_0**

There is no statistically significant evidence that the population mean is different from 70.

2. INDEPENDENT t-TEST

Suppose we don't have one class. Suppose we have two different classes. Can we use the same test?
No.

Previously, we were comparing **one group with a known/claimed value**.
Now we have:
Class A vs Class B
These are **two different groups of students**.
Therefore, we use:
Independent t-test
An independent t-test is used to compare the means of **two different, unrelated groups**.

For example:

- Class A vs Class B
- Boys vs Girls
- Online class vs Offline class
- Teaching Method 1 vs Teaching Method 2

The important point is:
The people in Group A are different from the people in Group B.

Problem

Suppose a teacher wants to know:

"Is there a significant difference between the average marks of Class A and Class B?"

We have:

Class A

60, 65, 70, 75, 80

Class B

55, 60, 62, 68, 70

Are these the same students?

No.

Are we measuring the same student twice?

No.

We have two separate groups.

Therefore:

Two different groups

↓

Independent t-test

Why Independent t-test?

We want to compare:

Average of Class A

↓

?

↓

Average of Class B

But just looking at the averages is not enough.

Suppose:

Class A average = 70

Class B average = 63

There is a difference:

70 - 63 = 7 marks

But the question is:

Is this 7-mark difference large enough to be considered statistically significant, or could it have occurred because of random variation?

That is why we perform the **independent t-test**.

Formulate Hypotheses

Step 1: H_0

Null Hypothesis

There is **no significant difference** between the average marks of Class A and Class B.

Mathematically:

$$H_0: \mu_A = \mu_B$$

Simple meaning:

Both classes have the same population mean.

H_1

Alternative Hypothesis

There is a **significant difference** between the average marks of Class A and Class B.

$$H_1: \mu_A \neq \mu_B$$

Simple meaning:

The population means are different.

Because we are asking whether the means are **different in either direction**, this is a:

Two-tailed test

Class A could be higher

OR

Class B could be higher

So:

$$\mu_A \neq \mu_B$$

means **two-tailed**.

Significance Level

Set:

$$\alpha = 0.05$$

We are using a 5% significance level.

This gives us our decision rule:

p < 0.05 → Reject H_0

p > 0.05 → Fail to reject H_0

Calculate Mean of Class A

Class A:

60, 65, 70, 75, 80

Formula

$$\bar{x}_A = \frac{\sum x}{n}$$

Therefore:

$$\bar{x}_A = \frac{60 + 65 + 70 + 75 + 80}{5}$$

$$= \frac{350}{5}$$

$$\boxed{\bar{x}_A = 70}$$

The average mark of Class A is **70**.

Calculate Mean of Class B

Class B:

55, 60, 62, 68, 70

$$\bar{x}_B = \frac{55 + 60 + 62 + 68 + 70}{5}$$

$$= \frac{315}{5}$$

$$\boxed{\bar{x}_B = 63}$$

Therefore:

Class A mean = 70

Class B mean = 63

Observed difference:

$$70 - 63 = 7$$

So we observe a **7-mark difference**.

But remember:

We cannot decide only from the difference of 7.

We need the t-test.

Calculate Standard Deviation of Class A

Standard deviation tells us how much the students' marks vary around their class average.

For Class A:

60, 65, 70, 75, 80

Mean = 70.

Create the table:

Mark	Mark – Mean	Squared Difference
60	-10	100
65	-5	25
70	0	0
75	5	25
80	10	100
		250

Now calculate sample variance:

$$s_A^2 = \frac{250}{5-1}$$
$$= \frac{250}{4}$$
$$= 62.5$$

Standard deviation:

$$s_A = \sqrt{62.5}$$

sA≈7.91

Calculate Standard Deviation of Class B

Class B:
55, 60, 62, 68, 70
Mean = 63.

Mark	Mark – Mean	Squared Difference
55	-8	64
60	-3	9
62	-1	1
68	5	25
70	7	49
		148

Sample variance:

$$s_B^2 = \frac{148}{5-1}$$
$$= \frac{148}{4}$$
$$= 37$$

Standard deviation:

$$s_B = \sqrt{37}$$

$s_B \approx 6.08$

So now we have:

	Class A	Class B
Mean	70	63
SD	7.91	6.08
n	5	5

Independent t-test Formula

pooled/equal-variance independent t-test:

$$t = \frac{\bar{x}_A - \bar{x}_B}{s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

sp is the **pooled standard deviation**. It combines the variation from both groups.

First calculate pooled variance:

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$$

Substitute:

$$s_p^2 = \frac{(5-1)(62.5) + (5-1)(37)}{5+5-2}$$

$$= \frac{4(62.5) + 4(37)}{8}$$

$$= \frac{250 + 148}{8}$$

$$= \frac{398}{8}$$

$$= 49.75$$

Therefore:

$$s_p = \sqrt{49.75}$$

sp≈7.05

Calculate t-value

Now substitute everything into the t formula:

$$t = \frac{70 - 63}{7.05\sqrt{\frac{1}{5} + \frac{1}{5}}}$$

First:

$$70 - 63 = 7$$

And:

$$\frac{1}{5} + \frac{1}{5} = 0.4$$

$$\sqrt{0.4} \approx 0.632$$

Therefore:

$$t = \frac{7}{7.05 \times 0.632}$$

$$t \approx \frac{7}{4.46}$$

$t \approx 1.57$

What does 1.57 mean?

The observed difference between the two class means, relative to the variation within the two classes, gives us a t-value of approximately **1.57**.

Degrees of Freedom

For the pooled independent t-test:

$$df = n_A + n_B - 2$$

Here:

$$df = 5 + 5 - 2$$

$$\boxed{df = 8}$$

So:

$t \approx 1.57$

$df = 8$

Find the critical t- value

We have df=8 and significance level =0.05

T critical value is 2.306

Critical t-values = − 2.306 and +2.306

Compare with t and critical t- value

|t|= 1.57

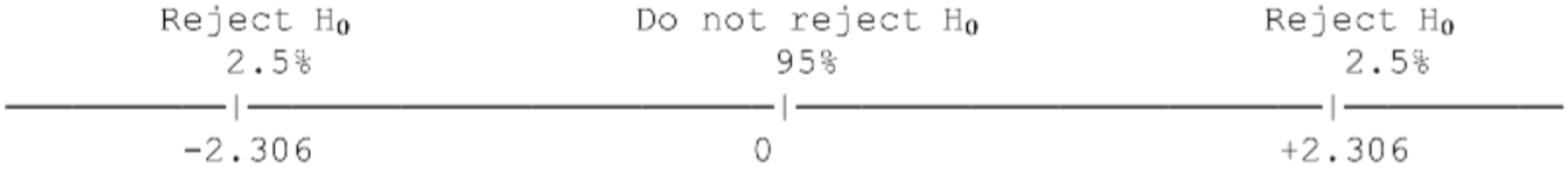
1.57< 2.306

Fail to reject H_0

Why are there two critical values?

Because we are checking for a difference in **either direction**.

The rejection regions look like this:



So:

$$t < -2.306$$

or

$$t > 2.306$$

→ **Reject** H_0

If:

$$-2.306 \leq t \leq 2.306$$

→ **Fail to reject H0**

Find p-value

Using:

$t \approx 1.57$

$df = 8$

two-tailed test

The p-value is approximately:

$$\boxed{p \approx 0.16}$$

Now compare:

$p = 0.16$

$\alpha = 0.05$

Since:

$$0.16 > 0.05$$

Therefore:

Fail to reject H0

Independent t-test — Detailed p-value Calculation

We have already calculated:

Class A

$$60,\ 65,\ 70,\ 75,\ 80$$

Mean:

$$\bar{X}_A = 70$$

Class B

$$55,\ 60,\ 62,\ 68,\ 70$$

Mean:

$$\bar{X}_B = 63$$

Therefore, the observed difference between the means is:

$$70 - 63 = 7$$

After calculating the standard deviations and pooled standard deviation, we obtained:

$$\boxed{t = 1.57}$$

and:

$$\boxed{df = 8}$$

Our hypotheses are:

$$H_0: \mu_A = \mu_B$$
$$H_1: \mu_A \neq \mu_B$$

and:

$$\alpha = 0.05$$

Because H_1 contains $\neq$, this is a **two-tailed test**.

Step 1 — What is a p-value?

The **p-value** tells us:

If the null hypothesis H_0 is true, how unusual is our observed result?

In our case:

$$H_0: \mu_A = \mu_B$$

So we temporarily assume:

The two classes actually have the same population mean.

Then we ask:

If the two population means are really equal, how unusual would it be to obtain a t-value as far from 0 as 1.57?

That probability is the **p-value**.

Step 2 — Why does the t-value matter?

We calculated:

$$t = 1.57$$

Remember that the t-value tells us how far our observed difference is from the null hypothesis, measured in standard-error units.

The center of the t-distribution is:

$$t = 0$$

If:

$$t = 0$$

there is essentially no standardized difference.

Our result is:

$$t = 1.57$$

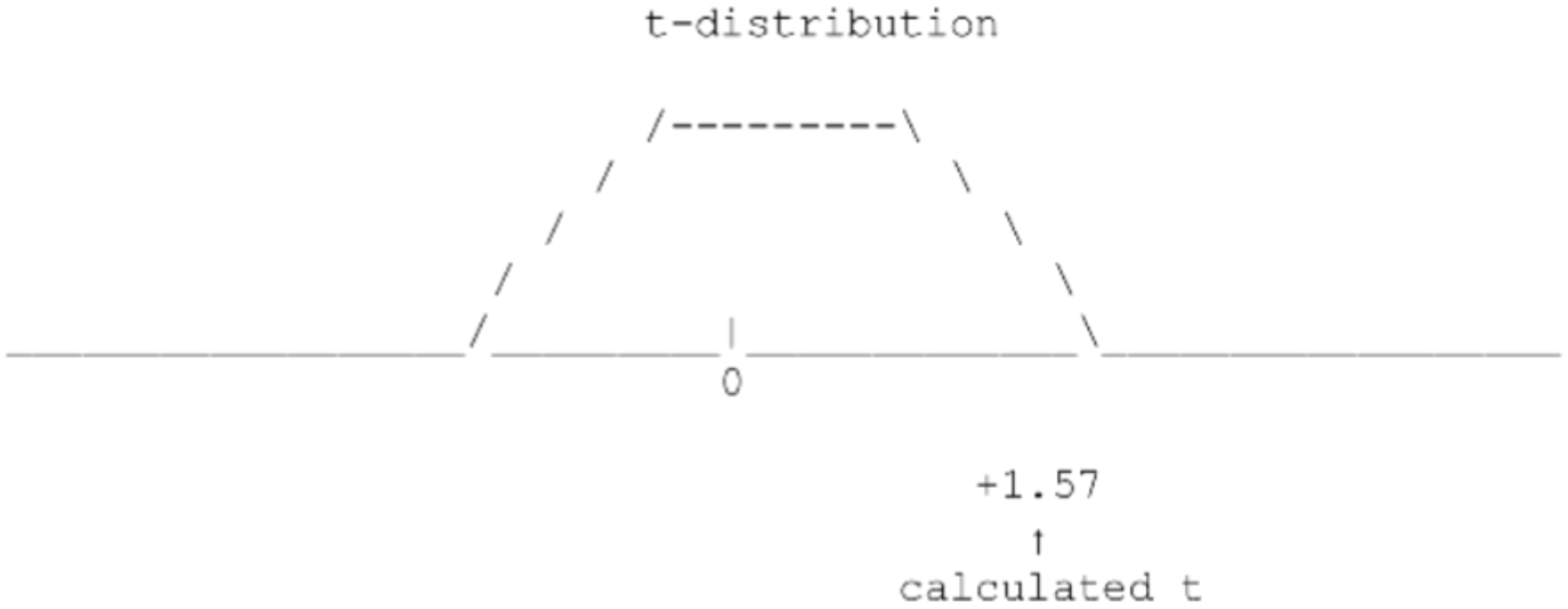
So, our result is **1.57 units away from the center**.

We therefore want to know:

How much area of the t-distribution lies beyond $+1.57$?

Step 3 — Draw the t-distribution

For $df = 8$, imagine the t-distribution like this:



The center is:

$$0$$

Our calculated value is:

$$+1.57$$

We are interested in the area **to the right of 1.57**.
That is called the **right-tail probability**.

Step 4 - Find the right-tail probability

We need:

$$P(T > 1.57)$$

where:

$$df = 8$$

This means:

For a t-distribution with 8 degrees of freedom, what is the probability of getting a t-value greater than 1.57?

Using a t-distribution calculation:

$$P(T > 1.57) \approx 0.078$$

So the **right-tail area** is approximately:

0.078

Where does 1.57 fall?

Look at:

$$1.397 < 1.57 < 1.860$$

So our t-value lies **between** the critical values for:

$$\alpha = 0.20$$

and

$$\alpha = 0.10$$

Therefore, from this standard table we can say:

$$\boxed{0.10 < p < 0.20}$$

Remember: these are **two-tailed probabilities**.

Step 5 — But our test is two-tailed

This is extremely important.

Our alternative hypothesis is:

$$H_1 : \mu_A \neq \mu_B$$

The symbol:

$$\neq$$

means:

We are interested in a difference in **either direction**.

The difference could be:

Class A greater than Class B

$$\mu_A > \mu_B$$

or:

Class A less than Class B

$$\mu_A < \mu_B$$

Therefore, we cannot look only at the right side.

We must also look at the **left side**.

Step 6 — Find the left-tail area

Our observed t-value is:

$$+1.57$$

Because the t-distribution is symmetric:

$$P(T < -1.57)$$

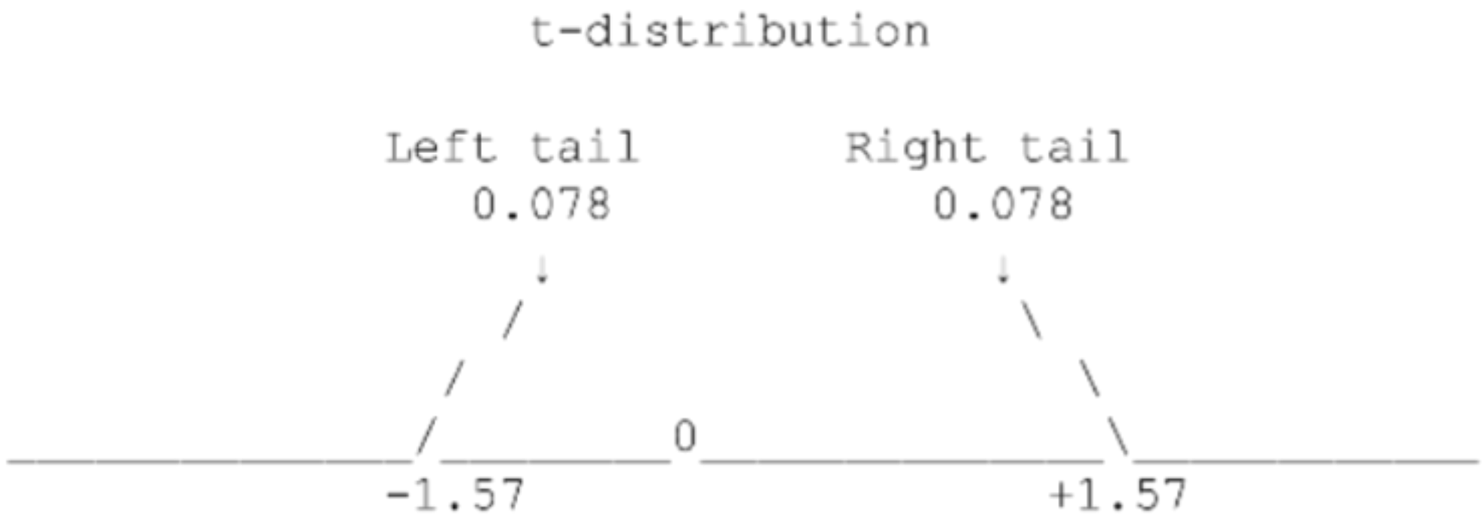
has the same area as:

$$P(T > 1.57)$$

Therefore:

$$P(T < -1.57) \approx 0.078$$

So we have:



Step 7 — Add the two tail areas

Since this is a **two-tailed test**, the p-value is the area in **both tails**.

Therefore:

$$p = 0.078 + 0.078$$
$$p = 0.156$$

Rounded to two decimal places:

$$\boxed{p \approx 0.16}$$

This is where your 0.16 comes from.

It is **not an arbitrary value**.

It comes from:

$$\boxed{\text{Right-tail } area + Left\text{-}tail\ area}$$

$$\boxed{0.078 + 0.078 = 0.156 \approx 0.16}$$

Step 8 — Why don't we just use 0.078?

Because 0.078 represents only **one tail**.

Your test is:

$$H_1: \mu_A \neq \mu_B$$

which is two-tailed.

Therefore:

$$\boxed{p = 2(0.078)}$$
$$\boxed{p = 0.156}$$

p≈0.16

Final Conclusion
Decision
p > 0.05, 0.16>0.05
→ **Fail to reject H_0**

Conclusion
There is **not enough statistical evidence** to conclude that the average marks of Class A and Class B are significantly different.
Important:
Do **not** say:
"H_0 is proved true."

Instead say:
"We fail to reject H_0 ."
Because our test has not provided enough evidence against H_0 .

3. Paired test

1. First understand when we use a paired t-test

A paired t-test is used when:

The same subjects are measured twice.

For example:

- marks **before training**
- marks **after training**

The same students appear in both measurements.

Example

Student	Before Training	After Training
1	60	70
2	65	72
3	70	75
4	62	68
5	68	74

Here, Student 1's before mark is compared with **Student 1's after mark**.
Therefore, the observations are **paired**.

Problem statement

A teacher wants to know whether a training program significantly changed students' marks.
The marks of the same 5 students were recorded:

Student	Before	After
1	60	70
2	65	72
3	70	75
4	62	68
5	68	74

Question
Did the training program significantly change the students' marks?
We will use a **paired t-test**.

Why can't we simply compare the two averages?

First calculate the two averages.

Before average

$$\bar{X}_{before} = \frac{60 + 65 + 70 + 62 + 68}{5}$$

Add the marks:

$$60 + 65 + 70 + 62 + 68 = 325$$

Therefore:

$$\bar{X}_{before} = \frac{325}{5}$$

$$\boxed{\bar{X}_{before} = 65}$$

After average

$$\bar{X}_{after} = \frac{70 + 72 + 75 + 68 + 74}{5}$$

Add:

$$70 + 72 + 75 + 68 + 74 = 359$$

Therefore:

$$\bar{X}_{after} = \frac{359}{5}$$

$$\boxed{\bar{X}_{after} = 71.8}$$

That's what the paired t-test determines.

So the average increased from:

$$65 \rightarrow 71.8$$

The difference is:

$$71.8 - 65 = 6.8$$

So marks increased by **6.8 marks on average**.

But we still need to ask:

Is this 6.8-mark increase large enough to be considered statistically significant?

State the hypotheses

For a paired t-test, it is easiest to work with the **differences** between the two measurements.

Let:

$$D = Before - After$$

Our hypotheses are:

Null hypothesis

$$\boxed{H_0 : \mu_D = 0}$$

Meaning:

There is no significant average change in marks.

Alternative hypothesis

$$\boxed{H_1 : \mu_D \neq 0}$$

Meaning:

There is a significant average change in marks.

Why is this a two-tailed test?

Look at:

$$H_1: \mu_D \neq 0$$

The symbol is:

$$\neq$$

This means we are looking for a change in **either direction**.

The training could:

- increase marks, or
- decrease marks.

Therefore:

Two-tailed *test*

And we use:

α=0.05

Step 1 — Calculate the differences

Now we create a new column:

$$D = Before - After$$

Calculate each student's difference.

Student 1

$$D = 60 - 70 = -10$$

Student 2

$$D = 65 - 72 = -7$$

Student 3

$$D = 70 - 75 = -5$$

Student 4

$$D = 62 - 68 = -6$$

Student 5

$$D = 68 - 74 = -6$$

So:

Student	Before	After	$D = Before - After$
1	60	70	-10
2	65	72	-7
3	70	75	-5
4	62	68	-6
5	68	74	-6

Therefore:

$$\boxed{D = [-10, -7, -5, -6, -6]}$$

Why are all the differences negative?

Students may ask this.

We defined:

$$D = Before - After$$

For example:

$$60 - 70 = -10$$

because the after mark is higher.

So, the negative value simply tells us:

After-training marks were higher than before-training marks.

It does **not** mean something is wrong.

If we had defined:

$$D = After - Before$$

then our differences would be:

$$10,7,5,6,6$$

and the t-value would be positive instead of negative.

The final two-tailed decision would be the same.

Step 2 — Calculate the mean difference

We have:

$$D = -10, -7, -5, -6, -6$$

First add them:

$$-10 + (-7) + (-5) + (-6) + (-6)$$
$$= -34$$

Number of pairs:

$$n = 5$$

Therefore:

$$\bar{D} = \frac{-34}{5}$$

$$\boxed{\bar{D} = -6.8}$$

This means:

On average, $Before - After = -6.8$.

Or, in easier words:

After-training marks were 6.8 marks higher on average.

Step 3 — Calculate deviations from the mean difference

Now we need the sample standard deviation of the differences.

We have:

$$\bar{D} = -6.8$$

For every difference, calculate:

$$D - \bar{D}$$

Student 1

$$-10 - (-6.8)$$
$$= -10 + 6.8$$
$$= -3.2$$

Student 2

$$-7 - (-6.8)$$
$$= -0.2$$

Student 3

$$-5 - (-6.8)$$
$$= 1.8$$

Student 4

$$-6 - (-6.8)$$
$$= 0.8$$

Student 5

$$-6 - (-6.8)$$

=0.8

So:

Student	Difference D	$D - \bar{D}$
1	-10	-3.2
2	-7	-0.2
3	-5	1.8
4	-6	0.8
5	-6	0.8

Step 4 — Square the deviations

Why do we square them?

Because some deviations are negative and some are positive.

If we simply added them, they would cancel each other.

So we calculate:

$$(D - \bar{D})^2$$

Student 1

$$(-3.2)^2 = 10.24$$

Student 2

$$(-0.2)^2 = 0.04$$

Student 3

$$(1.8)^2 = 3.24$$

Student 4

$$(0.8)^2 = 0.64$$

Student 5

$$(0.8)^2 = 0.64$$

Therefore:

Student	D	$D - \bar{D}$	$(D - \bar{D})^2$
1	-10	-3.2	10.24
2	-7	-0.2	0.04
3	-5	1.8	3.24
4	-6	0.8	0.64
5	-6	0.8	0.64

Now add the squared deviations:

$$10.24 + 0.04 + 3.24 + 0.64 + 0.64$$

$$= 14.80$$

Therefore:

$$\boxed{\sum (D - \bar{D})^2 = 14.80}$$

Step 5 — Calculate variance of the differences

Because these are **sample data**, we use:

$$s_D^2 = \frac{\sum (D - \bar{D})^2}{n - 1}$$

We have:

$$\sum (D - \bar{D})^2 = 14.80$$

and:

$$n = 5$$

Therefore:

$$s_D^2 = \frac{14.80}{5 - 1}$$

$$= \frac{14.80}{4}$$

sD2=3.70

Step 6 — Calculate standard deviation of differences

Standard deviation is the square root of variance.

Therefore:

$$s_D = \sqrt{3.70}$$

$$\boxed{s_D \approx 1.92}$$

So the standard deviation of the differences is approximately:

$$\boxed{1.92}$$

.Step 7 — Calculate the standard error

Now we need the **standard error of the mean difference.**

The formula is:

$$SE_{\bar{D}} = \frac{s_D}{\sqrt{n}}$$

We have:

$$s_D = 1.92$$

and:

$$n = 5$$

Therefore:

$$SE_{\bar{D}} = \frac{1.92}{\sqrt{5}}$$

We know:

$$\sqrt{5} \approx 2.236$$

Therefore:

$$SE_{\bar{D}} = \frac{1.92}{2.236}$$

$$\boxed{SE_{\bar{D}} \approx 0.86}$$

Using the unrounded standard deviation gives approximately:

$SED^{-}=0.860$

Step 8 — Calculate the t-statistic

For a paired t-test:

$$\boxed{t = \frac{\bar{D} - \mu_{D0}}{s_D/\sqrt{n}}}$$

Under the null hypothesis:

$$H_0: \mu_D = 0$$

Therefore:

$$\mu_{D0} = 0$$

We already know:

$$\bar{D} = -6.8$$

and:

$$SE_{\bar{D}} \approx 0.86$$

Therefore:

$$t = \frac{-6.8 - 0}{0.86}$$

$$t \approx -7.90$$

So:

t≈−7.90

Why is the t-value negative?

We calculated:

$$D = Before - After$$

Since After is generally larger:

$$Before - After < 0$$

Therefore:

$$\bar{D} = -6.8$$

which gives:

$$t = -7.90$$

The negative sign tells us the **direction of change**.

It indicates:

Before marks were generally lower than After marks.

If instead we calculated:

$$D = After - Before$$

we would get:

$$t = +7.90$$

For a **two-tailed test**, the magnitude is what matters for the decision:

|t|=7.90

Step 9 — Calculate degrees of freedom

For a paired t-test:

$$\boxed{df = n - 1}$$

We have 5 pairs.

Therefore:

$$df = 5 - 1$$

$$\boxed{df = 4}$$

Notice that we have **5 pairs**, not 10 independent observations.

That's why:

df=5−1=4

Step 10 — Find the critical t-value

We have:

$$df = 4$$

and:

$$\alpha = 0.05$$

Our alternative hypothesis is:

$$H_1: \mu_D \neq 0$$

Because we have ≠, this is a **two-tailed test**.

Therefore:

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

From the t-table:

$$\boxed{t_{critical} = \pm 2.776}$$

So the rejection regions are:

$$t < -2.776$$

or:

t>2.776

Step 11 — Compare calculated t with critical t

Our calculated value is:

$$t = -7.90$$

Critical values are:

$$-2.776, +2.776$$

Compare the absolute values:

$$| t | = 7.90$$

and:

$$t_{critical} = 2.776$$

Therefore:

$$7.90 > 2.776$$

So the calculated t-value lies in the **rejection region**.

Therefore:

$$\boxed{\text{Reject } H_0}$$

Step 12 — Now calculate the p-value

This is where we need to understand the p-value carefully.

We have:

$$t = -7.90$$

and:

$$df = 4$$

Because this is a two-tailed test, we look at both tails.

First take the absolute value:

$$| t | = 7.90$$

We want:

$$P(T < -7.90) + P(T > 7.90)$$

Because the t-distribution is symmetric:

$$P(T < -7.90) = P(T > 7.90)$$

The one-tail area is approximately:

$$0.000693$$

Therefore the two-tailed p-value is:

$$p = 2(0.000693)$$

$$p \approx 0.001386$$

Therefore:

$$\boxed{p \approx 0.0014}$$

So you can write:

$$\boxed{p \approx 0.0014}$$

or approximately:

$p<0.01$

Visual understanding of the p-value

Our t-value is:

$$-7.90$$

That's extremely far to the left of zero.

The t-distribution looks conceptually like:



The areas beyond $-7.90$and $+7.90$are extremely small.

Therefore:

$$p \approx 0.0014$$

This is much smaller than:

α=0.05

Step 13 — Compare p-value with α

We have:

$$p \approx 0.0014$$

and:

$$\alpha = 0.05$$

Compare:

$$0.0014 < 0.05$$

Therefore:

Reject H0

Final conclusion

Our hypotheses were:

$$H_0 : \mu_D = 0$$
$$H_1 : \mu_D \neq 0$$

We rejected H_0.

Therefore:

There is statistically significant evidence that students' marks changed after the training program.

Since the average mark increased from:

$$65$$

to:

$$71.8$$

we can additionally say:

In this sample, the marks increased after training.

Test	What are we comparing?	Example
One-sample t-test	One group vs a known value	Marks vs 70
Independent t-test	Two different groups	Class A vs Class B
Paired t-test	Same group twice	Before vs After training

And for the paired test:
The important trick is: don't treat Before and After as two unrelated groups. First calculate the difference for each student. Then perform the t-test on those differences.
That is the central idea of the **paired t-test**.

Example

Suppose we want to know:

“Did training improve students' marks?”

We measure the **same students before and after training**.

Student	Before	After
1	60	70
2	65	72
3	70	75
4	62	68
5	68	74

But the question is:
Is this increase statistically significant?

Statistically significant means:
The difference we observe is unlikely to be caused only by random chance.

Simple example
Suppose:
- Class A average = **75**
- Class B average = **70**

Difference = **5 marks**

Now, just seeing a 5-mark difference doesn't tell us whether the difference is meaningful.
A statistical test asks:
"Is this difference probably a real difference, or could it have happened randomly?"

If the test says the difference is **statistically significant**, it means there is enough statistical evidence to say:
"This difference is unlikely to be just due to chance."

We cannot simply look at the numbers.

So, we perform a **t-test**.
Because these are the **same students before and after**, we use a:
Paired t-test

Now apply H_0 and H_1 specifically to the **t-test.**

- **Null Hypothesis**

  H_0: *There is no significant change in marks*

- **Alternative Hypothesis**

H1:There is a significant change in marks

```
Same students
     ↓
Before and After marks
     ↓
Numerical data
     ↓
Compare the two related means
     ↓
Paired t-test
```

Complete Demonstration of t-test

Problem Statement

- A teacher wants to know whether a training program has significantly changed students' marks.
- We have marks of the **same students before and after training**.
- Here, the same students are measured twice, we use a **Paired t-test**.

```python
!pip install scipy
from scipy.stats import ttest_rel
# ----------------------------------------
# Paired t-test - Student Performance
# ----------------------------------------

# Step 1: Import required library
from scipy.stats import ttest_rel

# Step 2: Create marks before training
before = [60, 65, 70, 62, 68]

# Step 3: Create marks after training
after = [70, 72, 75, 68, 74]

# Step 4: Define the hypotheses
# H0: There is no significant change in marks
# H1: There is a significant change in marks

# Step 5: Perform Paired t-test
t_statistic, p_value = ttest_rel(before, after)

# Step 6: Display the results
print("Paired t-test")
print("----------------------")
print("t-statistic:", t_statistic)
print("p-value:", p_value)

# Step 7: Set significance level
alpha = 0.05
```

```python
# Step 8: Make the decision
if p_value < alpha:
    print("Decision: Reject H0")
    print("Conclusion: There is a significant change in students' marks.")
else:
    print("Decision: Fail to reject H0")
    print("Conclusion: There is not enough evidence to conclude that students' marks changed significantly.")
```

OUTPUT:

```
Paired t-test
----------------------
t-statistic: -7.90483943457971
p-value: 0.0013855069102196623
Decision: Reject H0
Conclusion: There is a significant change in students' marks.
```

p-value	Decision
$p < 0.05$	Reject H_0
$p \geq 0.05$	Fail to reject H_0

- **t-statistic:** Measures **how large the difference between the two means is compared with the variation in the data**. (**How big is the difference?)**
- **p-value:** Tells us **whether the observed difference could have happened by chance?**. (**Is the difference statistically significant or could it)**

test_rel() is the **statistical test**.
The test calculates the **test statistic and p-value**.

```
Hypothesis Testing
        ↓
Choose t-test
        ↓
Perform t-test
        ↓
ttest_rel()
        ↓
Get p-value
        ↓
Make decision
```

What is a p-value?

The p-value tells us how strong the evidence is against the null hypothesis.

We normally use:

$$\alpha = 0.05$$

Decision rule

p-value	Decision
$p < 0.05$	Reject H_0
$p \geq 0.05$	Fail to reject H_0

For example:
p-value = 0.0013

0.0013 < 0.05

Therefore:
Reject H_0
Conclusion: There is statistically significant evidence that the students' marks changed after the training.

Chi-square Test

The t-test is suitable when we are working with numerical data and means. But what if our variables are categorical?

For example:
Is course preference related to gender?

Now our variables are:
- Gender → Male/Female
- Course → Python/Java/C++

These are **categorical variables**.

Therefore, instead of a t-test, we use the:
Chi-square test

What is Chi-square Test?

The chi-square test is a statistical test used to determine whether two categorical variables are associated with each other.

Example:

Gender	Python	Java	C++
Male	40	30	20
Female	50	20	10

Question:

Is course preference related to gender?

Hypotheses for Chi-square test

- **Null Hypothesis — H_0**

Gender and course preference are **not associated**.

In simple words:

There is no relationship between gender and course preference.

- **Alternative Hypothesis — H_1**

Gender and course preference **are associated**.

In simple words:

There is a relationship between gender and course preference.

Chi-Square Test: Student Course Preference

We have:

Gender	Python	Java	C++	Total
Male	40	30	20	90
Female	50	20	10	80
Total	**90**	**50**	**30**	**170**

Here:

- Rows = Gender
- Columns = Course Preference
- Data = **counts/frequencies**

We want to know:

Is course preference related to gender?

Step 1: State the hypotheses

H_0 — Null Hypothesis

Gender and course preference are independent.

In simple words:

Gender has no relationship with course preference.

H_1 — Alternative Hypothesis

Gender and course preference are associated.

In simple words:

Gender and course preference are related.

Take:

$\alpha = 0.05$

Step 2: Calculate Expected Frequencies

This is the most important step in the Chi-square test.

We have the observed table:

Gender	Python	Java	C++	Total
Male	40	30	20	90
Female	50	20	10	80
Total	90	50	30	170

The formula is:

$$Expected = \frac{Row\ Total \times Column\ Total}{Grand\ Total}$$

Expected value for Male–Python

Row total = 90

Column total = 90

Grand total = 170

Therefore:

$$E = \frac{90 \times 90}{170}$$

$$E = \frac{8100}{170}$$

$$\boxed{E = 47.65}$$

So although we **observed 40 males choosing Python**, if gender and course preference were independent, we would expect about **47.65**.

Expected value for Male–Java

$$E = \frac{90 \times 50}{170}$$

$$= \frac{4500}{170}$$

$$\boxed{26.47}$$

Expected value for Male–C++

$$E = \frac{90 \times 30}{170}$$

$$= \frac{2700}{170}$$

$$\boxed{15.88}$$

Expected value for Female–Python

$$E = \frac{80 \times 90}{170}$$

$$= \frac{7200}{170}$$

$$\boxed{42.35}$$

Expected value for Female–Java

$$E = \frac{80 \times 50}{170}$$

$$= \frac{4000}{170}$$

$$\boxed{23.53}$$

Expected value for Female–C++

$$E = \frac{80 \times 30}{170}$$

$$= \frac{2400}{170}$$

Step 3: Create the Expected Frequency Table
Now we have:

Gender	Python	Java	C++
Male	47.65	26.47	15.88
Female	42.35	23.53	14.12

These are the **expected frequencies**.

So we now have:

Observed

Gender	Python	Java	C++
Male	40	30	20
Female	50	20	10

Expected

Gender	Python	Java	C++
Male	47.65	26.47	15.88
Female	42.35	23.53	14.12

Step 4: Calculate Chi-square value

The formula is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where:

- O = Observed frequency
- E = Expected frequency

We calculate this for **every cell**.

Cell 1: Male–Python

Observed:

$$O = 40$$

Expected:

$$E = 47.65$$

Therefore:

$$\frac{(40 - 47.65)^2}{47.65} = \frac{(-7.65)^2}{47.65} = \frac{58.52}{47.65}$$

=1.228

Cell 2: Male–Java

$$\frac{(30 - 26.47)^2}{26.47} = \frac{3.53^2}{26.47} = \frac{12.46}{26.47}$$

$$\boxed{0.471}$$

Cell 3: Male–C++

$$\frac{(20 - 15.88)^2}{15.88} = \frac{4.12^2}{15.88}$$

=1.069

Cell 4: Female–Python

$$\frac{(50-42.35)^2}{42.35}$$
$$=\frac{7.65^2}{42.35}$$
$$\boxed{1.382}$$

Cell 5: Female–Java

$$\frac{(20-23.53)^2}{23.53}$$
$$=\frac{(-3.53)^2}{23.53}$$
$$\boxed{0.529}$$

Cell 6: Female–C++

$$\frac{(10-14.12)^2}{14.12}$$
$$=\frac{(-4.12)^2}{14.12}$$

=1.201

Step 5: Add all the values

Now add all six contributions:

$$\chi^2 = 1.228 + 0.471 + 1.069 + 1.382 + 0.529 + 1.201$$

Therefore:

$$\boxed{\chi^2 \approx 5.88}$$

This is the **Chi-square statistic**.

So your Python code:

```
chi2, p_value, degrees_of_freedom, expected = chi2_contingency(data)
```

will give approximately:

```
Chi-square statistic: 5.8765
```

Step 6: Calculate Degrees of Freedom

Formula:

$$\boxed{df = (Rows - 1)(Columns - 1)}$$

We have:

- Rows = 2 → Male, Female
- Columns = 3 → Python, Java, C++

Therefore:

$$df = (2 - 1)(3 - 1)$$
$$df = 1 \times 2$$

df=2

Step 7: Find the critical value from Chi-square table

We have:

$$df = 2$$

and:

$$\alpha = 0.05$$

From the Chi-square table:

$$\boxed{\chi^2_{critical} = 5.991}$$

Now compare:

$$\chi^2_{calculated} = 5.877$$

with:

$$\chi^2_{critical} = 5.991$$

Therefore:

$$5.877 < 5.991$$

So we **fail to reject** H_0.

Degrees of freedom (*df*)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

Step 8: Using the p-value method

The exact p-value for:

$$\chi^2 = 5.8765, df = 2$$

is approximately:

$$\boxed{p = 0.05296}$$

or approximately:

$$\boxed{p = 0.053}$$

Now compare:

$$p = 0.053$$

with:

$$\alpha = 0.05$$

Since:

$$0.053 > 0.05$$

we:

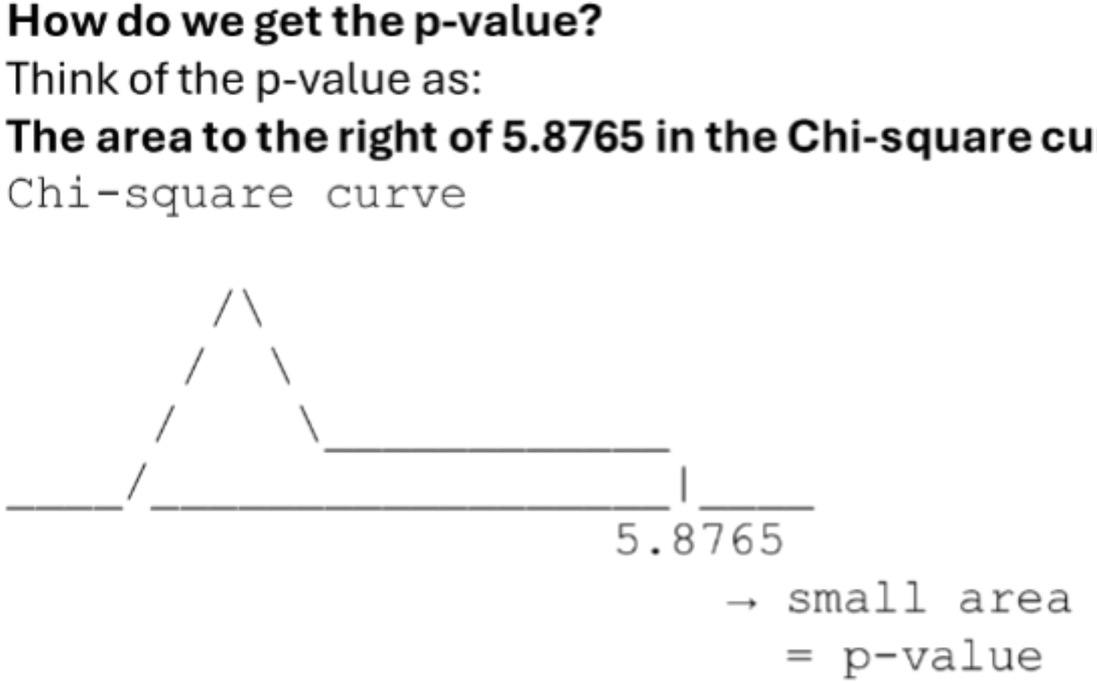
Fail to reject H0

Conclusion: There is not enough evidence to say that **gender and course preference are associated**.

How do we get the p-value?

Think of the p-value as:

The area to the right of 5.8765 in the Chi-square curve.



For $df = 2$, the exact calculation gives:

$$p \approx 0.05296$$

So:

p=0.053

Can we get 0.053 from the Chi-square table?

Not exactly.

From the table for $df = 2$:

- At $p = 0.10$, critical value = **4.605**
- At $p = 0.05$, critical value = **5.991**

Our value:

$$5.8765$$

is between them:

$$4.605 < 5.8765 < 5.991$$

Therefore:

$$\boxed{0.05 < p < 0.10}$$

The **exact 0.05296** comes from calculating the area using the Chi-square distribution (or software).

Step 9: Final conclusion

Statistical conclusion

Since the p-value (0.053) is greater than 0.05, we fail to reject the null hypothesis.

In simple words

There is **not enough statistical evidence at the 5% significance level** to conclude that gender and course preference are associated.

Notice that the p-value is **very close to 0.05**, but because it is slightly greater than 0.05, the decision at the 5% level is **fail to reject** H_0.

Complete Demonstration of Chi-square Test

"In the previous example, our data was numerical and we were comparing marks. What if our data is categorical?"

Problem Statement
Suppose we want to know:
Is course preference related to gender?

Our data is:

Gender	Python	Java	C++
Male	40	30	20
Female	50	20	10

Both variables are categorical:

- Gender → Male/Female
- Course → Python/Java/C++

Therefore, we use a **Chi-square test of independence**.

```python
# ----------------------------------------
# Chi-square Test - Student Course Preference
# ----------------------------------------

# Step 1: Import required library
from scipy.stats import chi2_contingency

# Step 2: Create observed frequency table
data = [
    [40, 30, 20],   # Male
    [50, 20, 10]    # Female
]

# Step 3: Define the hypotheses
# H0: Gender and course preference are independent
# H1: Gender and course preference are associated

# Step 4: Perform Chi-square test
chi2, p_value, degrees_of_freedom, expected = chi2_contingency(data)

# Step 5: Display the results
print("Chi-square Test")
print("----------------------")
print("Chi-square statistic:", chi2)
print("p-value:", p_value)
print("Degrees of freedom:", degrees_of_freedom)

# Step 6: Display expected frequencies
print("\nExpected Frequencies:")
print(expected)

# Step 7: Set significance level
alpha = 0.05
```

```python
# Step 8: Make the decision
if p_value < alpha:
    print("\nDecision: Reject H0")
    print("Conclusion: Gender and course preference are associated.")
else:
    print("\nDecision: Fail to reject H0")
    print("Conclusion: There is not enough evidence to conclude that gender and course preference are associated.")
```

- Chi-square value → How much do observed and expected values differ?
- P – value tells us whether the observed difference is **statistically significant**.
    - **degree of freedom** represents the number of values that are **free to vary** in the contingency table.
      For a simple Chi-Square table:
      degrees of freedom = (number of rows − 1) × (number of columns − 1)
    - expected gives the **expected frequencies**.

```
Chi-square Test
----------------------
Chi-square statistic: 5.876543209876544
p-value: 0.05295718061255369
Degrees of freedom: 2

Expected Frequencies:
[[47.64705882 26.47058824 15.88235294]
 [42.35294118 23.52941176 14.11764706]]

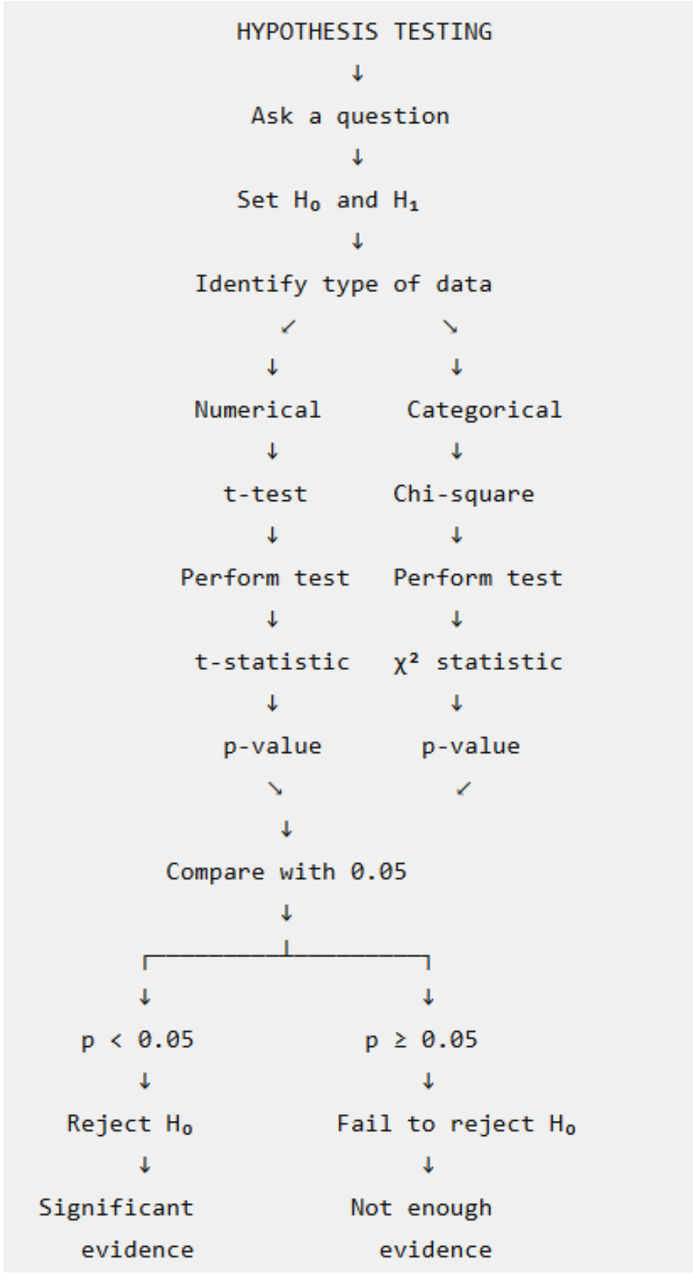
Decision: Fail to reject H0
Conclusion: There is not enough evidence to conclude that gender and course preference are associated.
```

Variable	Simple meaning
`chi2`	Difference between observed and expected
`p_value`	Is the difference significant?
`degrees_of_freedom`	Number of independent values
`expected`	What we expect if there is **no association**

Difference between Paired t-test and Chi-square test

	Paired t-test	Chi-square test
Data type	Numerical	Categorical
Main purpose	Compare means	Check association
Example	Before vs After marks	Gender vs Course
Python function	`ttest_rel()`	`chi2_contingency()`
Main output	t-statistic, p-value	χ^2 statistic, p-value
Decision	Compare p with 0.05	Compare p with 0.05

"Hypothesis testing is the overall process. A statistical test such as a t-test or chi-square test is the method we use to test the hypotheses. The test gives us a p-value, and the p-value helps us make the final decision about H_0 ."


HYPOTHESIS TESTING
Ask a question
Set H_0 and H_1
Identify type of data
Numerical
Categorical
t-test
Chi-square
Perform test
Perform test
t-statistic
χ^2 statistic
p-value
p-value
Compare with 0.05
$p < 0.05$
$p \geq 0.05$
Reject H_0
Fail to reject H_0
Significant evidence
Not enough evidence

Data Visualization Techniques

1. What is Data Visualization?

Data visualization is the process of representing data using visual elements such as graphs, charts, plots, and colors so that the data can be understood easily.

(OR)

Data visualization means converting data into a visual form so that we can understand the data more easily.

(OR)

Data visualization means converting numbers or data into a visual form that is easier to understand.

Why do we need it?
Suppose we have the following marks:
45, 50, 52, 60, 65, 70, 72, 75, 80, 90

If I simply give these numbers to you people and ask:
"What can you understand from this data?"
You are going to look at every value and it is difficult to understand.

Instead, if I represent the same data using a graph,

we can immediately see:

- where most marks are concentrated
- the lowest and highest marks
- how the marks are distributed
- whether any value is unusual
- whether two variables are related

So, visualization makes **data easier to understand and interpret**.

Why is Data Visualization Used?

Data visualization is mainly used to **find patterns, trends, comparisons, relationships, and unusual values in data**.

Example

Imagine a college has marks of **1,000 students**.
Looking at 1,000 rows in an Excel sheet is difficult.

But if we create a graph, we can quickly answer questions such as:

- How many students scored above 80?
- Which mark range has the most students?
- Are there any extremely low or high marks?
- Is attendance related to marks?
- Which subject has better performance?

Therefore:
Visualization converts complex data into information that can be understood quickly.

Why is Data Visualization Important?

1. Makes data easy to understand

Large amounts of numerical data can be difficult to read and understand.
If you people use the graph, it gives us a **visual summary to understand the data**.
Example:
Instead of reading:
45, 50, 52, 60, 65, 70, 72, 75, 80, 90
we can look at a graph and understand the data distribution immediately.

2. Helps identify patterns

Visualization can show patterns that may not be obvious from raw data.
For example:
Students who study more hours generally get higher marks.
A scatter plot can make this pattern visible.

A pattern means something that happens in a recognizable way.
Example:
Study Hours → Marks

1 hour → 40
2 hours → 45
3 hours → 50
4 hours → 60
5 hours → 65
6 hours → 72
7 hours → 80
8 hours → 88

We can notice that marks generally increase as study hours increase.
A scatter plot makes this pattern much easier to see.

3. Helps identify trends

A **trend** tells us how something changes over time.
Example:

Month	Sales
January	100
February	120
March	150
April	180

A line graph will show that sales are increasing.

So, visualization helps us answer:
"Is the value increasing or decreasing over time?"

4. To compare values

Suppose we have:

Department	Average Marks
CSE	82
ECE	75
EEE	70
ME	68

A chart can immediately show which department has higher or lower performance.

5. To identify unusual values

Suppose student marks are:
40, 45, 50, 55, 60, 65, 70, 75, 150
Most marks are between 40 and 75.
But **150** is unusual.

This type of unusual observation is called an **outlier**.
A box plot is particularly useful for identifying such values.

Different Data Visualization Techniques are:

1. Histogram
2. Box Plots
3. Scatter Plots
4. Heatmap
5. Pair plots

1. Histogram

What is a Histogram?

A histogram is a graph used to show the distribution of numerical data by grouping values into intervals.

In very simple words:

A histogram tells us how the values are distributed.

Why do we use a Histogram?

Suppose we have the marks of 100 students.

42, 45, 48, 51, 52, 54, 55, ...

Looking at every individual mark is difficult.

Instead, we can group them:

40– 49

50– 59

60– 69

70– 79

80– 89

90– 99

Then count how many students fall into each range.

For example:

Marks	Number of Students
40– 49	2
50– 59	1
60– 69	2
70– 79	1
80– 89	2

The histogram represents these frequencies using bars.

What does it help us understand?

We can quickly identify:

- where most values occur
- where fewer values occur
- whether data is concentrated
- whether data is spread out
- whether the distribution is symmetric or skewed

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
df = pd.read_csv("students (1).csv")
```

```python
print("First 5 Records:")
print(df.head())
```

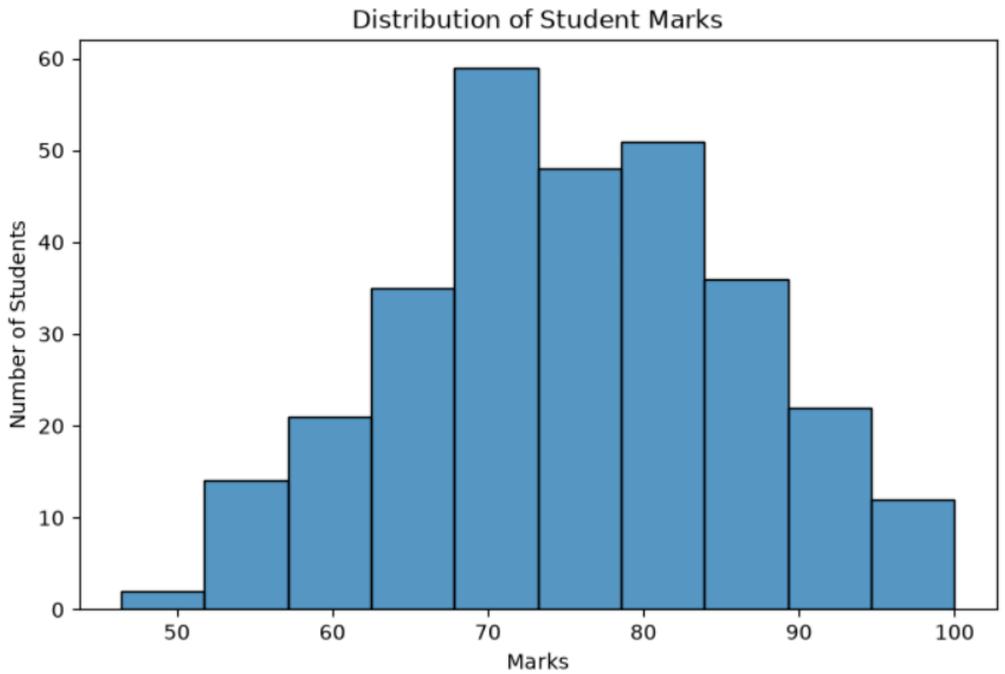
```
First 5 Records:
  StudentID  StudyHours  Attendance  PrevMarks  Marks Branch Result
0     S1000        5.08       82.32      65.44  82.46    ISE   Pass
1     S1001        6.16       68.27      82.55  79.14    CSD   Pass
2     S1002        5.42       98.16      67.71  76.89    CSE   Pass
3     S1003        2.93       79.50      68.25  62.16    ISE   Fail
4     S1004        8.01       93.33      60.55  86.49    ISE   Pass
```

```python
plt.figure(figsize=(8, 5)) # Histogram

sns.histplot(
    df["Marks"],
    bins=10
)

plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")

plt.show()
```



2. Box Plot

What is a Box Plot?

A **box plot** is a visualization technique used to summarize the distribution of numerical data.

It shows important statistical values such as:

- Minimum
- Q1
- Median
- Q3
- Maximum

It can also help identify **outliers**.

In simple words:

A box plot tells us how the data is spread and whether there are unusual values.

Why do we need a Box Plot?

Suppose we have:

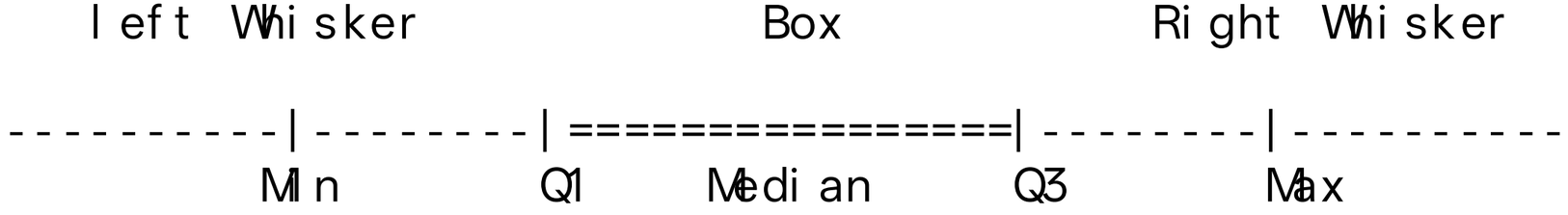
40, 45, 50, 60, 65, 72, 80, 88

We may want to know:

- What is the middle value?
- How widely are the marks spread?
- Where is the middle 50% of students?
- Is there an unusually high or low mark?

A box plot gives us this information in a compact form.

Important Parts of a Box Plot



- **Median**

The **median** is the middle value of the ordered data.
For example:
40, 45, 50, 60, 65, 72, 80, 88
There are 8 values, so the median is the average of the two middle values:
Median = (60 + 65) / 2
= 62.5

- **Q1**

Q1 is the first quartile.
It represents approximately the **25th percentile**.
It separates the lower 25% of the data from the remaining data.

- **Q3**

Q3 is the third quartile.
It represents approximately the **75th percentile**.
It separates the upper 25% of the data from the remaining data.

- **IQR**

The **Interquartile Range** tells us how spread out the middle 50% of the data is.
Formula:
IQR = Q3 − Q1
This is important for detecting outliers.

- **Outlier**

An **outlier** is an observation that is unusually far from the rest of the data.
For example:
40, 45, 50, 60, 65, 72, 80, 88, 150

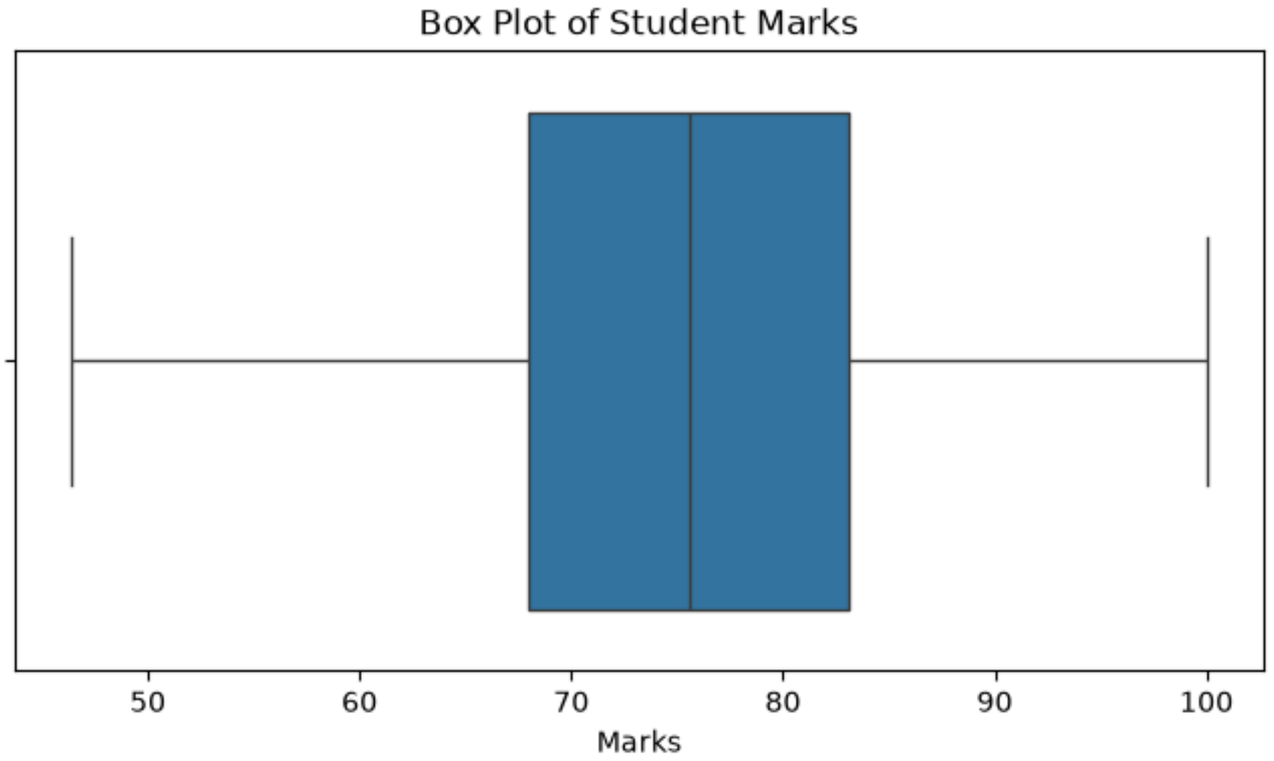
Here:
150 is much higher than the other values.
The box plot can show it separately.

```
plt.figure(figsize=(8, 4))  # Box Plot

sns.boxplot(
    x=df["Marks"]
)

plt.title("Box Plot of Student Marks")
plt.xlabel("Marks")

plt.show()
```

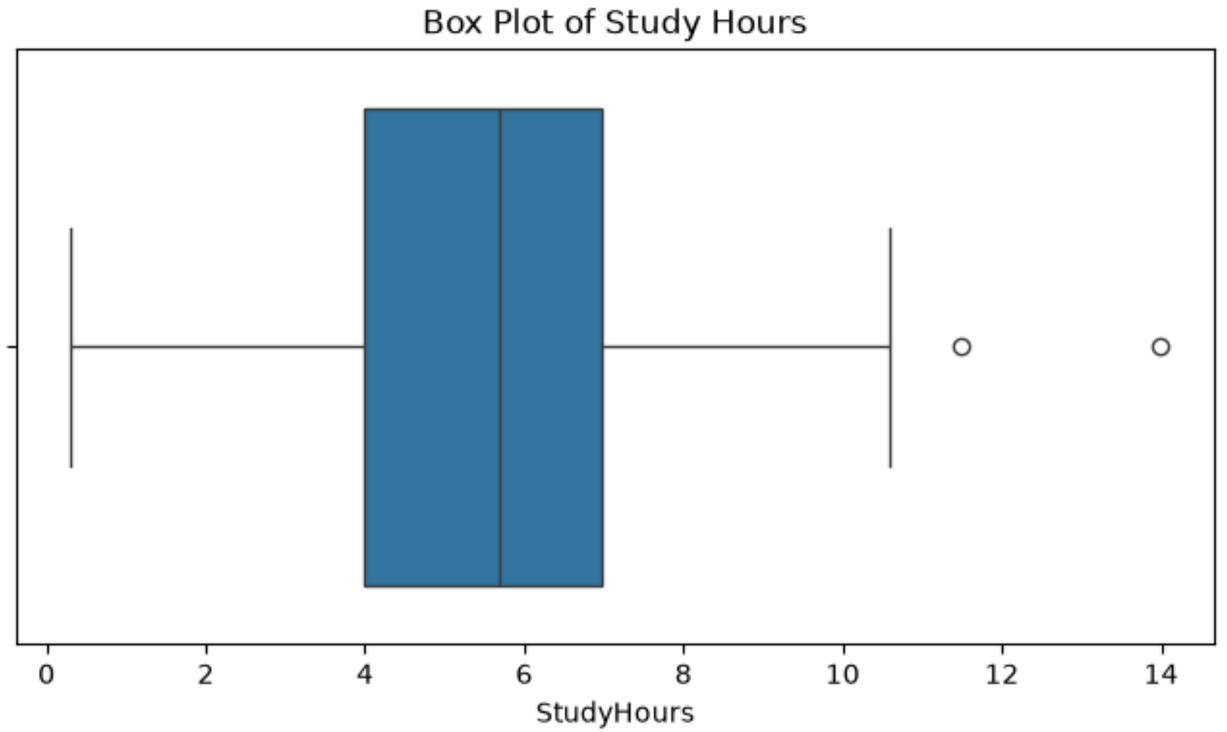


```
plt.figure(figsize=(8, 4))  # Box Plot

sns.boxplot(
    x=df["StudyHours"]
)

plt.title("Box Plot of Student Marks")
plt.xlabel("StudyHours")

plt.show()
```



3. Scatter Plot

What is a Scatter Plot?

A **scatter plot** is used to show the relationship between **two numerical variables**.

In simple words:

A scatter plot shows whether two variables are related to each other.

Here our two variables are:

Study Hours

Marks

Why do we use it?

Consider:

Study Hours	Marks
1	40
2	45
3	50
4	60
5	65
6	72
7	80
8	88

We want to answer:
"Do students who study more hours tend to get higher marks?"

A scatter plot helps us visually answer this.

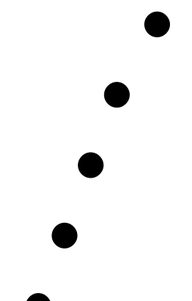
Each student becomes one point.

For example:
Student A → (1, 40)
Student B → (2, 45)
Student C → (3, 50)

The first number represents study hours.
The second number represents marks.

Understanding the pattern

If the points generally move upward:



we say there is a **positive relationship**.

That means:

As one variable increases, the other variable tends to increase.

Example

Study Hours ↑

↓

Marks generally ↑

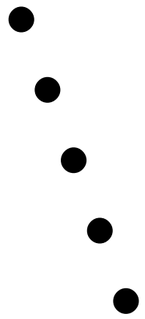
Negative relationship

Suppose we study:

Price vs Number of customers

As price increases, the number of customers may decrease.

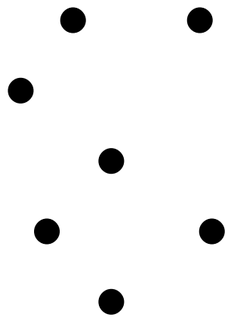
The points may look like:



This is a **negative relationship**.

No relationship

If the points are scattered randomly:



there may be **no clear relationship**.

```python
plt.figure(figsize=(8, 5))

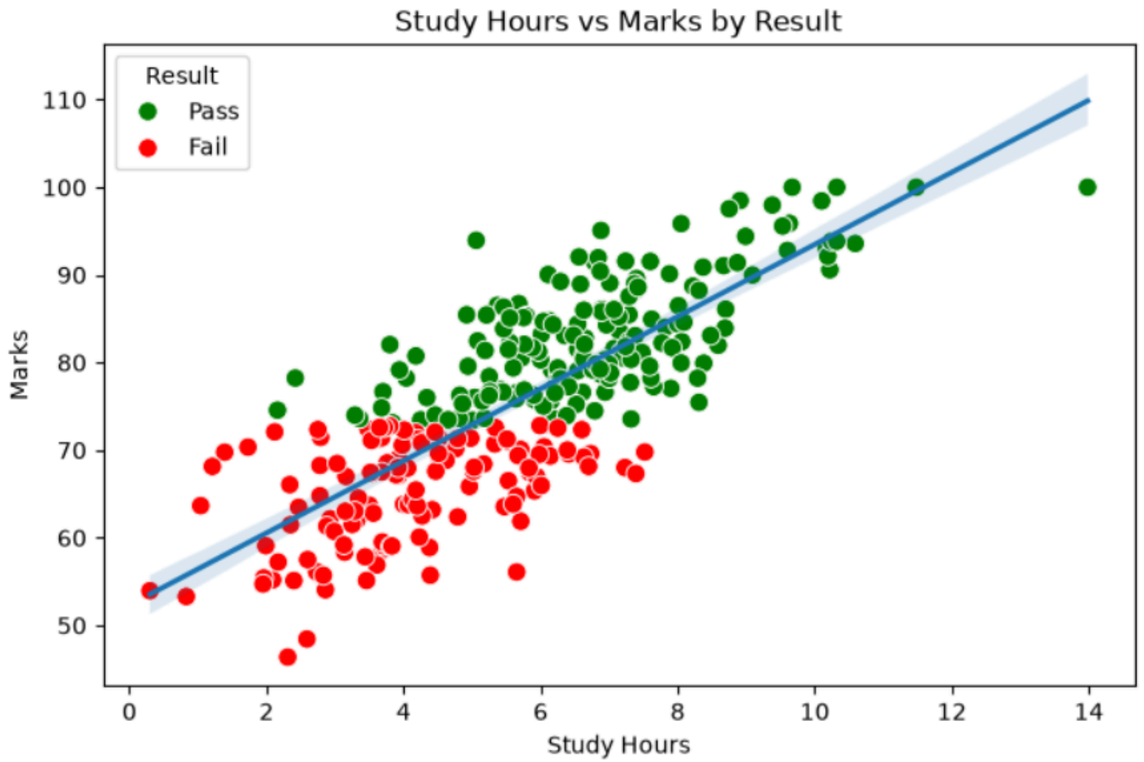
sns.scatterplot(
    x="StudyHours",
    y="Marks",
    data=df,
    hue="Result",
    palette={
        "Pass": "green",
        "Fail": "red"
    },
    s=70
)

# Add overall trend line
sns.regplot(
    x="StudyHours",
    y="Marks",
    data=df,
    scatter=False
)

plt.title("Study Hours vs Marks by Result")
plt.xlabel("Study Hours")
plt.ylabel("Marks")

plt.legend(title="Result")

plt.show()
```



4. Heatmap

What is a Heatmap?

A heatmap is a visualization technique that represents numerical values using different colors.

In simple words:

A heatmap converts numbers into colors so that high and low values can be identified quickly.

Why do we use a Heatmap?

Suppose we have marks of students in three subjects:

Student	Python	Maths	Statistics
A	90	85	88
B	70	65	72
C	45	50	48
D	80	75	82

If we have hundreds of students and many subjects, reading all the numbers becomes difficult.

A heatmap uses colors to make the pattern visible.

Conceptually:

	Python	Maths	Statistics
Student A	High	High	High
Student B	Mid	Mid	Mid
Student C	Low	Low	Low
Student D	High	High	High

The actual visualization uses a **color scale**.

For example:

```
Low -------------------------- High
  Dark                        Light
```

The exact colors depend on the chosen color map.

Heatmap for Correlation

Another very important use of a heatmap is **correlation analysis**.
Suppose we calculate the correlation between:

- Study Hours
- Attendance
- Marks

We may get:

	Study Hours	Attendance	Marks
Study Hours	1.00	0.70	0.85
Attendance	0.70	1.00	0.75
Marks	0.85	0.75	1.00

A heatmap can represent these values using colors.
This allows us to quickly identify:
Which variables have a strong relationship?
For example:
Study Hours ↔ Marks = 0.85
This indicates a strong positive relationship in this illustrative dataset.

```python
correlation = df[
    [
        "StudyHours",
        "Attendance",
        "PrevMarks",
        "Marks"
    ]
].corr()

print("\nCorrelation Matrix:")
print(correlation)
```

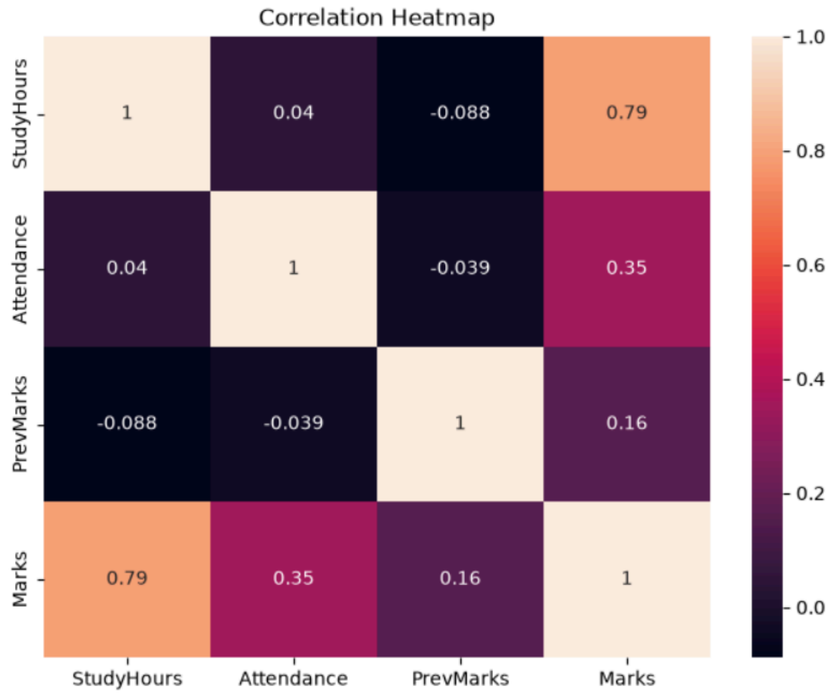
```
Correlation Matrix:
             StudyHours  Attendance  PrevMarks     Marks
StudyHours     1.000000    0.040008  -0.087901  0.790987
Attendance     0.040008    1.000000  -0.038883  0.354317
PrevMarks     -0.087901   -0.038883   1.000000  0.163115
Marks          0.790987    0.354317   0.163115  1.000000
```

```python
plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True
)

plt.title("Correlation Heatmap")

plt.show()
```

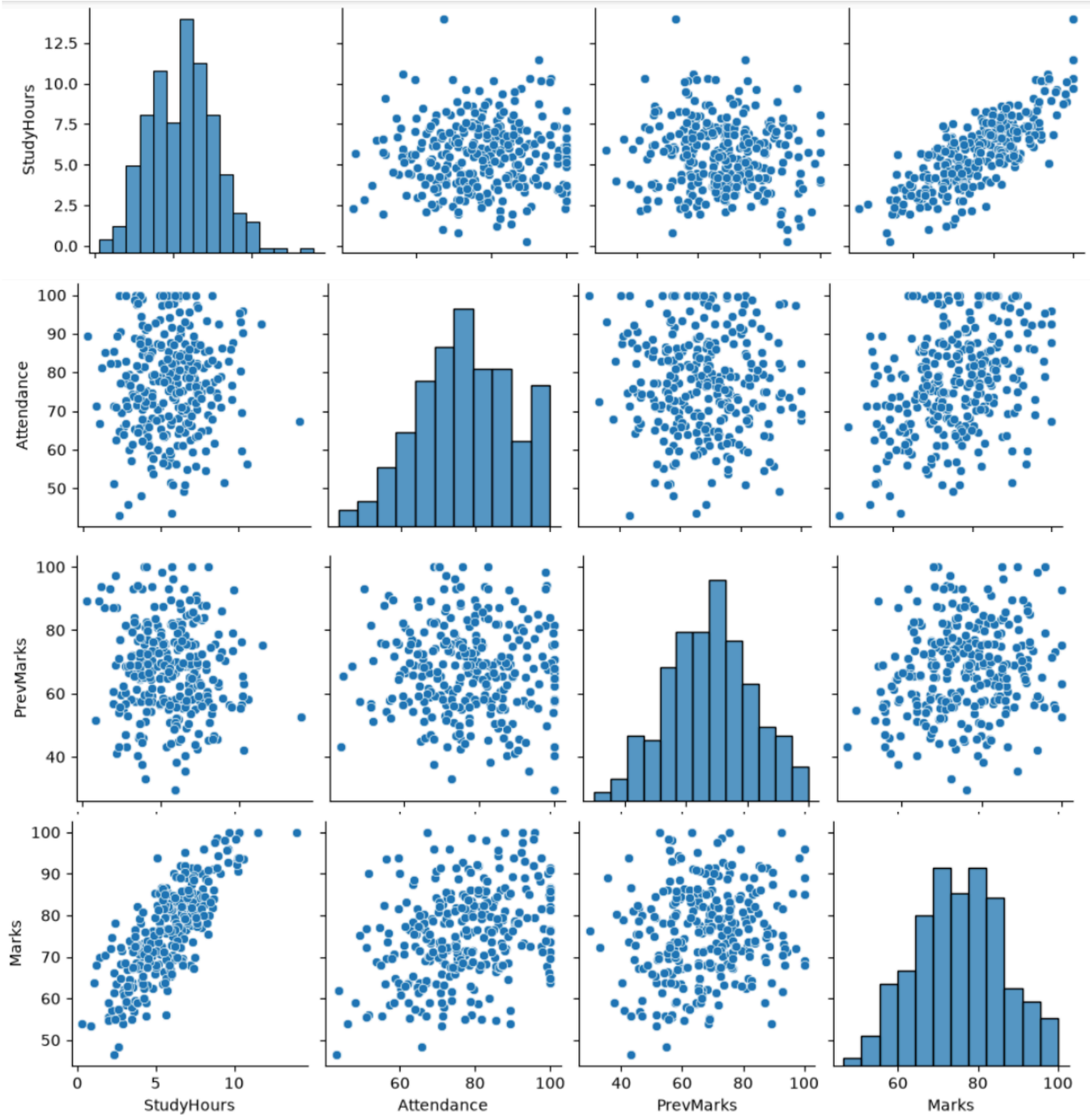


5. Pair Plots

It is used to examine the relationship between multiple numerical variables at the same time.

```
sns.pairplot(
    df[
        [
            "StudyHours",
            "Attendance",
            "PrevMarks",
            "Marks"
        ]
    ]
)

plt.show()
```


StudyHours
Attendance
PrevMarks
Marks
0.0
2.5
5.0
7.5
10.0
12.5
50
60
70
80
90
100
40
0
5
10

EDA using Pandas and Matplotlib/Seaborn

Problem Statement

"Suppose our college has a dataset containing student details, marks, attendance, and study hours. The principal asks us to understand the overall performance of students."

Student	Branch	Gender	Marks	Attendance	StudyHours
Asha	CSE	F	85	92	4
Ravi	ECE	M	72	80	3
Priya	CSE	F	91	95	5
Arun	ECE	M	65	70	2
Kiran	CSE	M	78	85	3
Divya	ECE	F	88	90	4
Rahul	CSE	M	55	60	1
Sneha	ECE	F	82	88	4
Vijay	CSE	M	74	78	2
Meena	ECE	F	95	98	5

"Can we understand the performance of all students just by looking at this table?"

no. **"We need a systematic way to explore this data. This process is called Exploratory Data Analysis, or EDA."**

First, we should **explore the data**.

We can ask:

- How many students are there?
- What columns are available?
- What is the average mark?
- Are there missing values?
- Are there duplicate records?
- What is the highest and lowest mark?
- How are marks distributed?
- Is attendance related to marks?
- Which branch has higher average marks?

This exploration is **EDA**.

What is EDA?

EDA = Exploratory Data Analysis

- **Exploratory** → We explore the data.
- **Data** → Information collected about students.
- **Analysis** → We examine and understand that information.

So:
EDA means exploring a dataset to understand what the data contains, identify problems, find patterns, and obtain useful insights.

(OR)

"EDA is like getting to know your dataset before making any decisions using it."

Here, we are going to learn **three different Libraries** and they are **Pandas, Matplotlib and Seaborn**
Before coding, we need to understand the role of each library.

Pandas

"Pandas helps us work with the data." (OR) **" Pandas is mainly used to work with and analyze tabular data."**

Think of Pandas as:
A tool for working with tables/dataframes.

It helps us:

- load data
- inspect data
- filter data
- calculate statistics
- find missing values
- remove duplicates
- group data

Matplotlib is a Python library used for data visualization.
Think of it as:
A tool for creating graphs.

It can create:
- line charts
- bar charts
- histograms
- scatter plots
- pie charts

Seaborn
Seaborn is another Python visualization library built on Matplotlib.
In simple words:
Seaborn makes statistical visualizations easier and provides convenient functions for analyzing relationships and distributions.

Seaborn is especially useful for:
- distributions
- relationships between variables
- box plots
- heatmaps
- categorical comparisons

Important Note:

Pandas → Understand the data
Matplotlib → Draw the data
Seaborn → Draw statistical patterns easily

Import the Libraries

Now open Google Colab / Jupyter Notebook.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Create the Dataset

```
data = {
    "Student": ["Asha", "Ravi", "Priya", "Arun", "Kiran",
                "Divya", "Rahul", "Sneha", "Vijay", "Meena"],

    "Branch": ["CSE", "ECE", "CSE", "ECE", "CSE",
               "ECE", "CSE", "ECE", "CSE", "ECE"],

    "Gender": ["F", "M", "F", "M", "M",
               "F", "M", "F", "M", "F"],

    "Marks": [85, 72, 91, 65, 78, 88, 55, 82, 74, 95],

    "Attendance": [92, 80, 95, 70, 85, 90, 60, 88, 78, 98],

    "StudyHours": [4, 3, 5, 2, 3, 4, 1, 4, 2, 5]
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

```
  Student Branch Gender  Marks  Attendance  StudyHours
0    Asha    CSE      F     85          92           4
1    Ravi    ECE      M     72          80           3
2   Priya    CSE      F     91          95           5
3    Arun    ECE      M     65          70           2
4   Kiran    CSE      M     78          85           3
5   Divya    ECE      F     88          90           4
6   Rahul    CSE      M     55          60           1
7   Sneha    ECE      F     82          88           4
8   Vijay    CSE      M     74          78           2
9   Meena    ECE      F     95          98           5
```

Display the Dataset

First EDA Question: What does our dataset look like?

Now begin actual EDA.

1. head()

df.head()

head() displays the first 5 rows by default.

Why?

"When we receive a large dataset, we don't want to display thousands of rows. We first look at a few rows to understand its structure."

We can also specify:
df.head(3)
This displays the first 3 rows.

```
df.head()
```

	Student	Branch	Gender	Marks	Attendance	StudyHours
0	Asha	CSE	F	85	92	4
1	Ravi	ECE	M	72	80	3
2	Priya	CSE	F	91	95	5
3	Arun	ECE	M	65	70	2
4	Kiran	CSE	M	78	85	3

Look at the Last Rows

```
df.tail()
```

	Student	Branch	Gender	Marks	Attendance	StudyHours
5	Divya	ECE	F	88	90	4
6	Rahul	CSE	M	55	60	1
7	Sneha	ECE	F	82	88	4
8	Vijay	CSE	M	74	78	2
9	Meena	ECE	F	95	98	5

2. Find Number of Rows and Columns

```
df.shape
```

```
(10, 6)
```

3. Find Column Names

```
df.columns
```

```
Index(['Student', 'Branch', 'Gender', 'Marks', 'Attendance', 'StudyHours'], dtype='str')
```

4. Check Data Types

```
df.dtypes
```

```
Student          str
Branch           str
Gender           str
Marks          int64
Attendance     int64
StudyHours     int64
dtype: object
```

5. Dataset Information

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Student     10 non-null     str
 1   Branch      10 non-null     str
 2   Gender      10 non-null     str
 3   Marks       10 non-null     int64
 4   Attendance  10 non-null     int64
 5   StudyHours  10 non-null     int64
dtypes: int64(3), str(3)
memory usage: 612.0 bytes
```

This is an important EDA command.

It tells us:

- number of rows
- column names
- number of non-null values
- data types
- memory usage

Non-Null Count

Suppose:
Marks 10 non-null
There are 10 rows and all 10 have marks.

If we get:
Marks 8 non-null

then:
"There are 10 students, but only 8 have marks. Therefore, 2 marks are missing."

6. Statistical Summary

```
df.describe()
```

	Marks	Attendance	StudyHours
count	10.000000	10.000000	10.000000
mean	78.500000	83.600000	3.300000
std	12.340088	11.815244	1.337494
min	55.000000	60.000000	1.000000
25%	72.500000	78.500000	2.250000
50%	80.000000	86.500000	3.500000
75%	87.250000	91.500000	4.000000
max	95.000000	98.000000	5.000000

count
Number of available numerical observations and it is also count non missing values
mean
Average value.
std
Standard deviation — how spread out the values are around the average.
min
Smallest value.
25%
25th percentile.
50%
Median.
75%
75th percentile.
max
Largest value.

7. Missing Values

"A dataset may not always be perfect. Some values may be missing." This produces True and False . But this is difficult to read for a large dataset.

```
df.isnull()
```

	Student	Branch	Gender	Marks	Attendance	StudyHours
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
5	False	False	False	False	False	False
6	False	False	False	False	False	False
7	False	False	False	False	False	False
8	False	False	False	False	False	False
9	False	False	False	False	False	False

```
df.isnull().sum()
```

```
Student       0
Branch        0
Gender        0
Marks         0
Attendance    0
StudyHours    0
dtype: int64
```

Therefore:
"There are no missing values in this dataset."

isnull() checks whether a value is missing.
sum() counts how many missing values are present in each column.

8. Duplicate Values

```
df.duplicated()
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
dtype: bool
```

This checks whether rows are duplicated.

To count duplicates:

```
df.duplicated().sum()
```

```
np.int64(0)
```

There are no duplicate rows.

Now comes to Visualization part

"So far, we have used numbers and tables to understand our dataset. But sometimes a graph can show a pattern much more clearly. "

Now move to **Matplotlib** and **Seaborn.**

1. Bar Chart Using Matplotlib

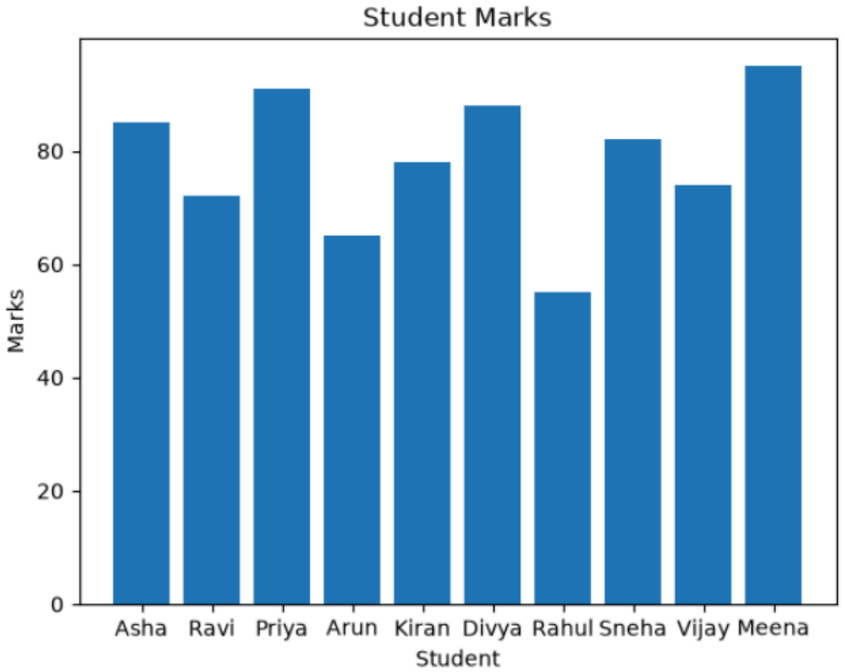
A **bar chart** is a visual tool that uses rectangular bars to represent and compare different categories of data. Bar charts are primarily used for **categorical data.**

Question:
Which student has the highest marks?

```
plt.bar(df["Student"], df["Marks"])

plt.xlabel("Student")
plt.ylabel("Marks")
plt.title("Student Marks")

plt.show()
```



Bar chart is mainly used to compare values across categories.

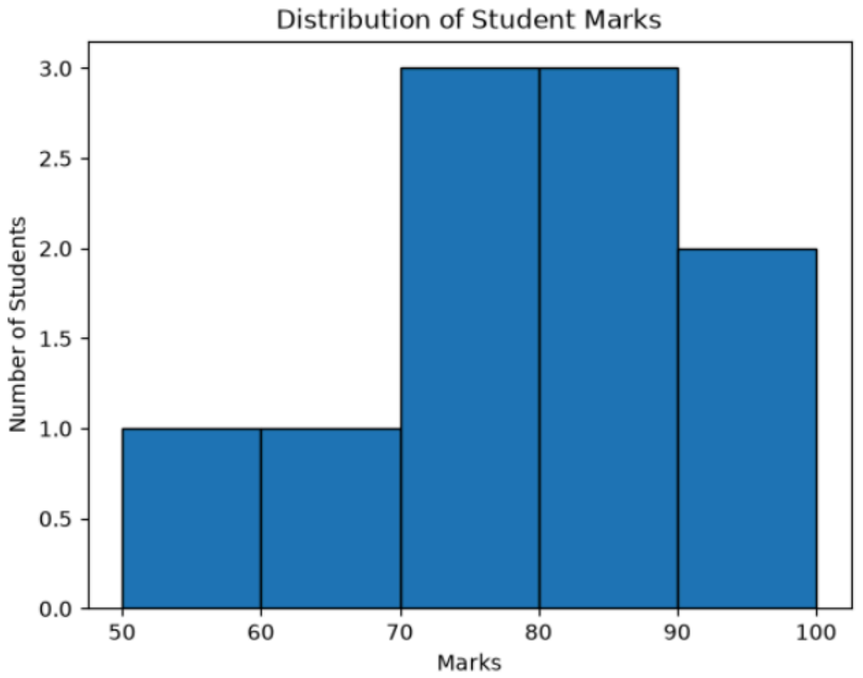
2. Histogram Using Matplotlib

"Instead of comparing individual students, what if we want to know how marks are distributed?"

```
plt.hist(
    df["Marks"],
    bins=[50, 60, 70, 80, 90, 100],
    edgecolor="black"
)

plt.xticks([50, 60, 70, 80, 90, 100])
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Student Marks")

plt.show()
```



Histogram → used to understand the distribution of numerical data.

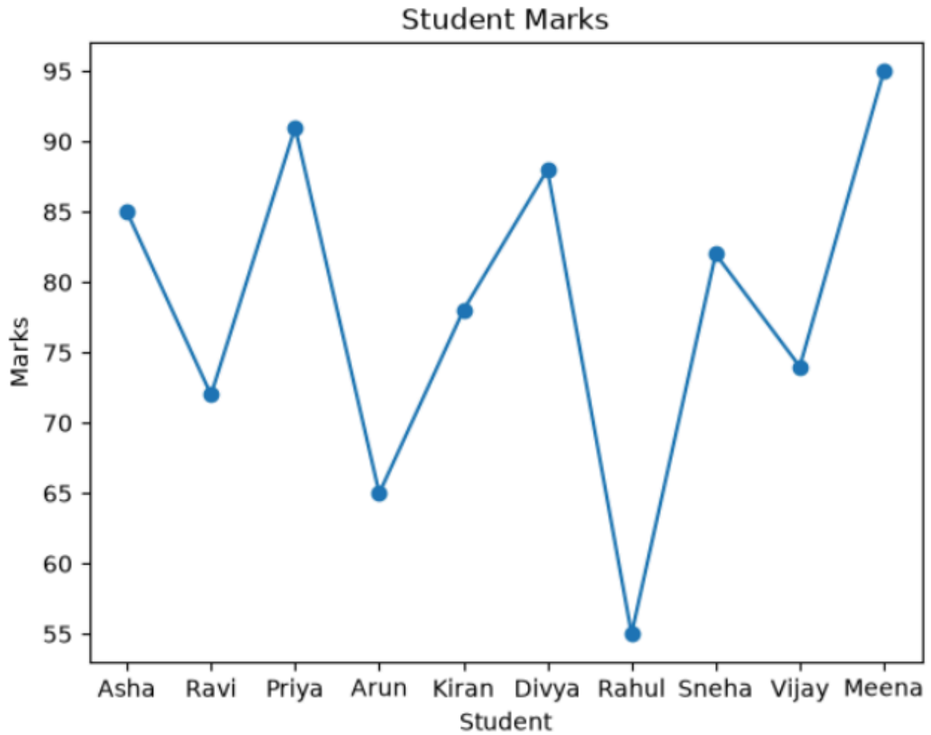
3. Line Plot using matplotlib

A line plot is a type of graph that represents data points using dots and connects those points with lines to show how a numerical value changes across an ordered sequence.

```
plt.plot(df["Student"], df["Marks"], marker="o")

plt.xlabel("Student")
plt.ylabel("Marks")
plt.title("Student Marks")

plt.show()
```



A line plot is used to show changes or trends in numerical data across an ordered sequence, especially over time.

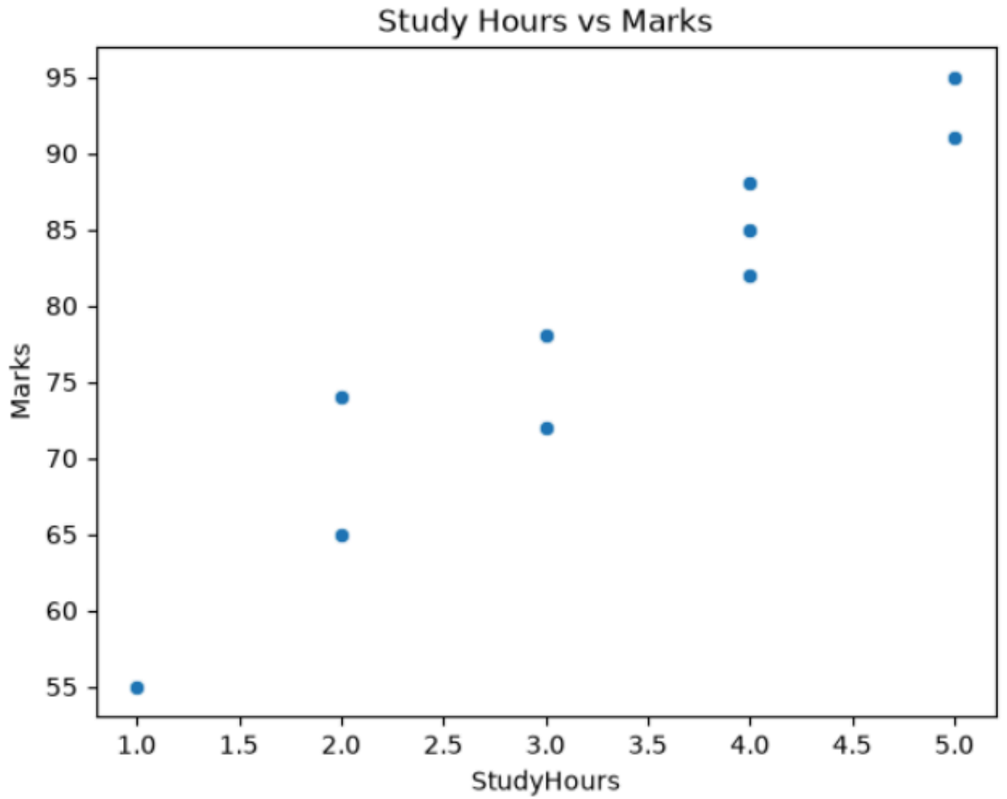
Example:
Month → Sales, Year → Population, Test Number → Marks, or Day → Temperature

4. Scatter Plot using Seaborn

"Is there any relationship between study hours and marks?" and It is used to see the relationship between two variables

```
sns.scatterplot(
    x="StudyHours",
    y="Marks",
    data=df
)

plt.title("Study Hours vs Marks")
plt.show()
```

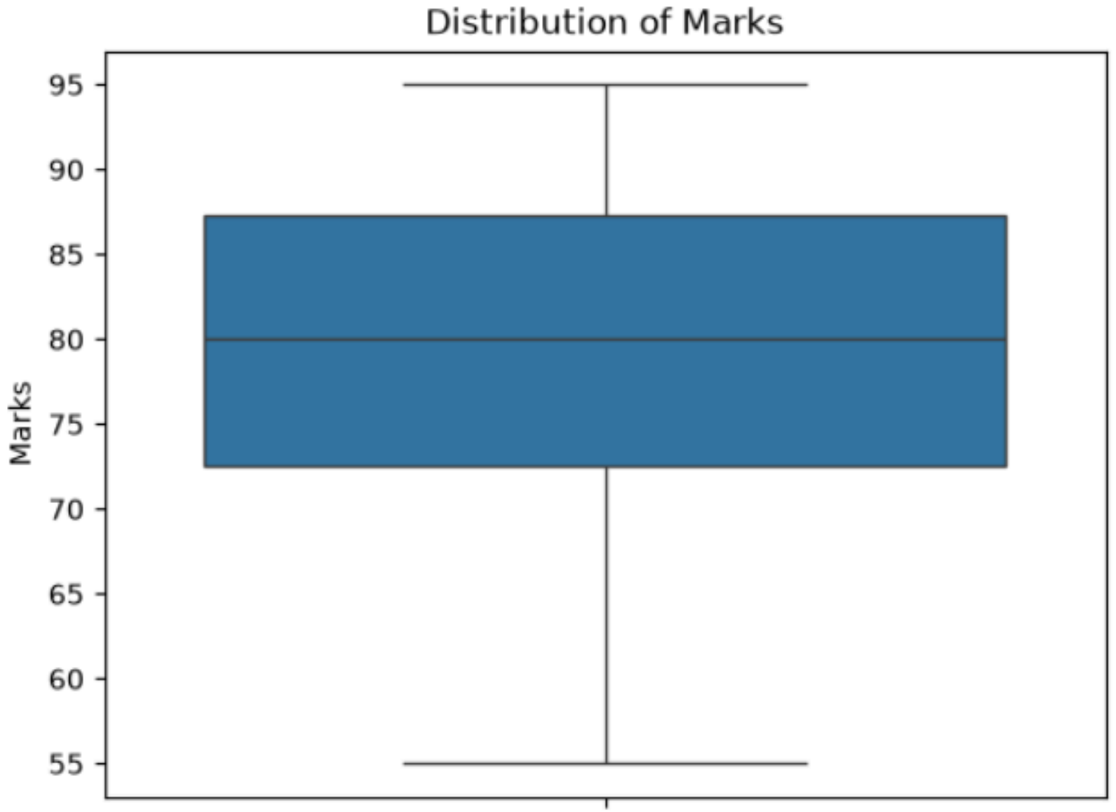


5. Box Plot using Seaborn

One of the most important EDA visualizations. It is used to see the spread of numerical data and identify the outliers.

```
sns.boxplot(y=df["Marks"])

plt.title("Distribution of Marks")
plt.show()
```



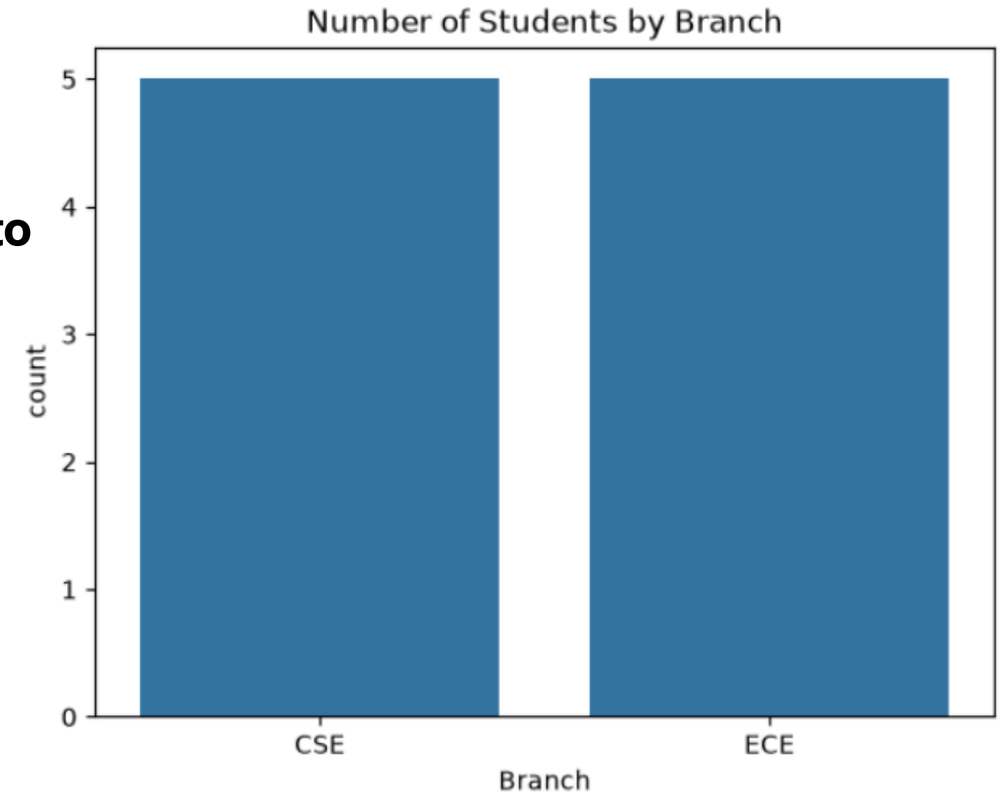
6. Count Plot Using Seaborn

Now analyze categorical data.

Question:
"How many students belong to each branch?"

```
sns.countplot(x="Branch", data=df)

plt.title("Number of Students by Branch")
plt.show()
```



Count plot counts how many observations belong to each category.

For example:
CSE → 5 students
ECE → 5 students

"A count simply means how many observations are present."

7. Correlation

To see relationships between numerical variables.

```
df[["Marks", "Attendance", "StudyHours"]].corr()
```

	Marks	Attendance	StudyHours
Marks	1.000000	0.988409	0.959318
Attendance	0.988409	1.000000	0.964667
StudyHours	0.959318	0.964667	1.000000

Correlation tells us how strongly two numerical variables move together.

The value generally ranges from:
-1 to +1

Basic interpretation

+1 → strong positive relationship
0 → little/no linear relationship
-1 → strong negative relationship

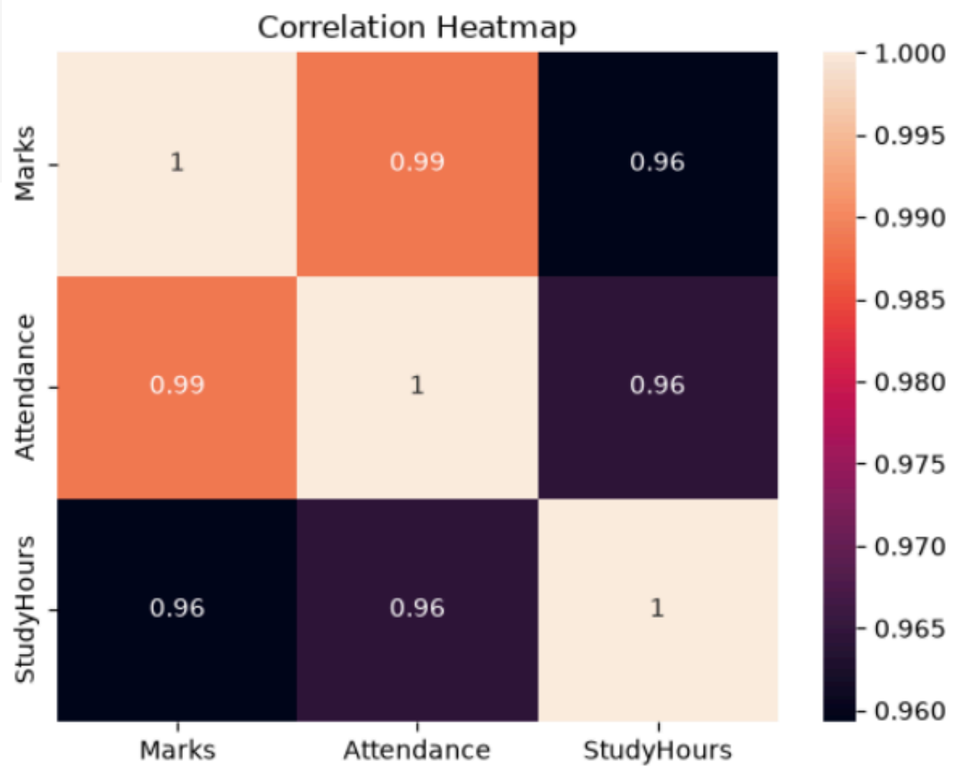
8. Correlation Heatmap using seaborn

Now visualize the correlation table.

```
correlation = df[["Marks", "Attendance", "StudyHours"]].corr()

sns.heatmap(
    correlation,
    annot=True
)

plt.title("Correlation Heatmap")
plt.show()
```



Important Note:

"EDA is like investigating a dataset. Pandas helps us inspect and analyze the data, while Matplotlib and Seaborn help us see the data visually. We explore the data, identify problems, discover patterns, and finally interpret what the data is telling us."

Univariate, Bivariate and Multivariate Analysis

What is Data Analysis?

Data analysis is the process of examining data to understand:

- What is happening?
- What patterns exist?
- Are variables related?
- What useful information can we obtain from the data?

When we perform analysis, the number of variables we study at the same time determines the type of analysis.

There are three important types:

1.Univariate Analysis
2.Bivariate Analysis
3.Multivariate Analysis

What is a Variable?

A **variable** is a characteristic or measurement that can have different values for different observations.

Example: Student Dataset

Student	Study Hours	Attendance	Assignments	Marks
A	2	65	3	48
B	3	70	4	55
C	4	75	5	62

Here:

- **Study Hours** is one variable.
- **Attendance** is one variable.
- **Assignments** is one variable.
- **Marks** is one variable.

The **Student** column identifies the observation.

Three Types of Analysis:

Type	Meaning	Number of Variables
Univariate	Analysis of one variable	1
Bivariate	Analysis of two variables	2
Multivariate	Analysis of three or more variables	3 or more

1. Univariate Analysis

Univariate analysis is the analysis of a single variable at a time.
The purpose is to understand the **distribution, central tendency, spread, or frequency** of that variable.

Example
Suppose we want to understand:
"How are the marks distributed among students?"
We consider only:
Marks
48, 55, 62, 70, 78, 85, 92
We don't consider Study Hours, Attendance, or Assignments.

Therefore:
One variable → Univariate Analysis

What Do We Study in Univariate Analysis?

For numerical data, we can investigate:

Measures of Central Tendency

- Mean
- Median
- Mode

These tell us about the **central or typical value**.

Measures of Dispersion

- Range
- Variance
- Standard deviation

These tell us **how spread out the data is**.

Other Analysis

- Minimum
- Maximum
- Percentiles
- Distribution
- Outliers

Common Visualizations

- Histogram
- Box plot
- Bar chart

Univariate Example

Suppose the marks are:
48, 55, 62, 70, 78, 85, 92
We can calculate:

[1]:
```
marks = [48, 55, 62, 70, 78, 85, 92]

import pandas as pd

df = pd.DataFrame({"Marks": marks})

print("Mean:", df["Marks"].mean())
print("Median:", df["Marks"].median())
print("Minimum:", df["Marks"].min())
print("Maximum:", df["Marks"].max())
print("Standard Deviation:", df["Marks"].std())
```

```
Mean: 70.0
Median: 70.0
Minimum: 48
Maximum: 92
Standard Deviation: 16.051998837112677
```

Visualization

Interpretation
We are trying to understand **only the Marks variable**.

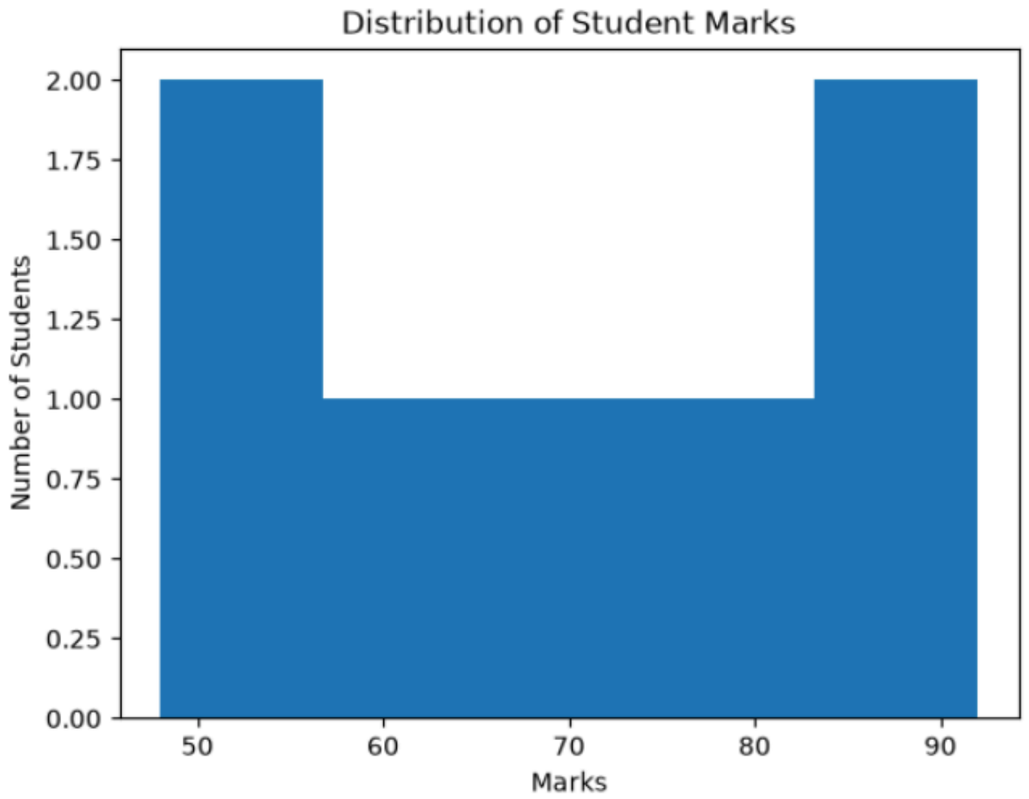
Therefore, this is:
Univariate Analysis

[2]:
```
import matplotlib.pyplot as plt

plt.hist(df["Marks"], bins=5)

plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Student Marks")

plt.show()
```



2. Bivariate Analysis

Bivariate analysis is the analysis of two variables together to understand whether there is a relationship or association between them.

Here we move from:
"What does one variable look like?"
to:
"How are two variables related?"

Bivariate Example
Consider:
Study Hours and **Marks**

Study Hours	Marks
2	48
3	55
4	62
5	70
6	78
7	85
8	92

Question:
"Is there a relationship between study hours and marks?"

Now we are investigating:
Study Hours + Marks

Therefore:
Two variables → Bivariate Analysis

Bivariate Analysis: What Can We Investigate?

Depending on the type of variables, we can study:

- Relationship
- Association
- Correlation
- Differences between groups
- Trends

Common visualizations

Numerical + Numerical

→ Scatter plot

Categorical + Numerical

→ Box plot / Bar chart

Categorical + Categorical

→ Cross-tabulation / Bar chart

Scatter Plot Example

[6]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Create the data
data = {
    "Study_Hours": [2, 3, 4, 5, 6, 7, 8],
    "Marks": [48, 55, 62, 70, 78, 85, 92]
}

# Create DataFrame
df = pd.DataFrame(data)

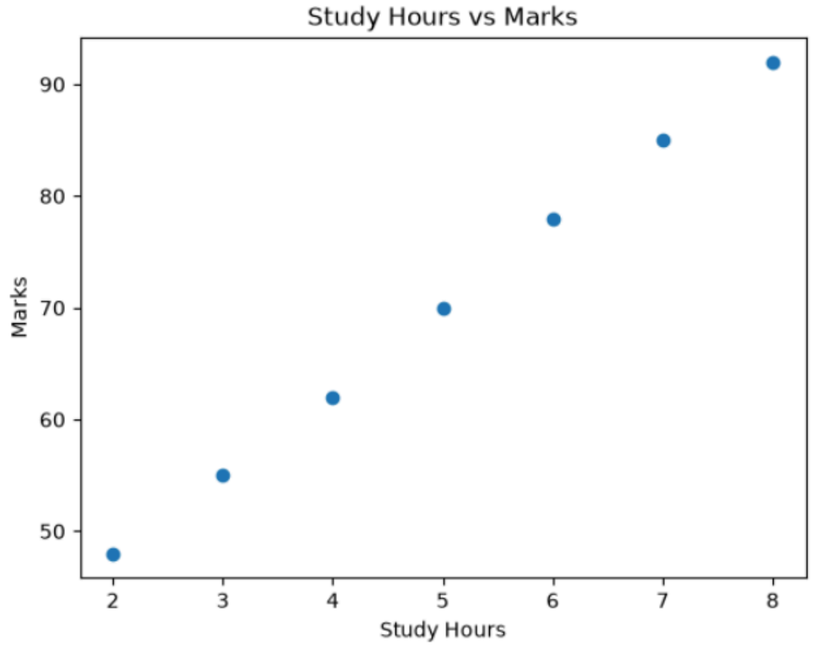
# Display the data
print(df)

# Create scatter plot
plt.scatter(df["Study_Hours"], df["Marks"])

plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")

plt.show()
```

```
   Study_Hours  Marks
0            2     48
1            3     55
2            4     62
3            5     70
4            6     78
5            7     85
6            8     92
```



The scatter plot helps us visually investigate whether the two variables have a relationship.
If the points generally move upward:

- There is a **positive relationship**.

If the points generally move downward:

- There is a **negative relationship**.

If there is no clear pattern:

- There may be little or no linear relationship.

Correlation Example

Correlation is a numerical measure used to describe the strength and direction of a linear relationship between two numerical variables.

Example:

```
[7]: correlation = df["Study_Hours"].corr(df["Marks"])

     print(correlation)

     0.9997227484813404
```

The correlation coefficient ranges from:
-1 to +1

Correlation	Meaning
Close to +1	Strong positive relationship
Close to -1	Strong negative relationship
Close to 0	Weak or no linear relationship

Important
Correlation shows association, not necessarily cause and effect.
For example, if study hours and marks are positively correlated, we should not automatically conclude that study hours alone caused the higher marks.

3. Multivariate Analysis

Multivariate analysis involves analyzing three or more variables together to understand relationships, patterns, or interactions among them.

Instead of looking at:
Study Hours → Marks
we can investigate:
Study Hours + Attendance + Assignments → Marks

Multivariate Example

Consider these variables:

- **Study Hours**
- **Attendance**
- **Assignments**
- **Marks**

Question:
"How are study hours, attendance, assignments and marks related to student performance?"
Now we are considering several variables simultaneously.

Therefore:
Three or more variables → Multivariate Analysis

Correlation Matrix

For multiple numerical variables, we can calculate correlations between every pair.

[10]:
```python
import pandas as pd

data = {
    "Study_Hours": [2, 3, 4, 5, 6, 7, 8],
    "Attendance": [65, 70, 75, 80, 85, 90, 95],
    "Assignments": [3, 4, 5, 6, 7, 8, 9],
    "Marks": [48, 55, 62, 70, 78, 85, 92]
}

df = pd.DataFrame(data)

# Calculate correlation
correlation = df[
    ["Study_Hours", "Attendance", "Assignments", "Marks"]
].corr()

print(correlation)
```

```
             Study_Hours  Attendance  Assignments     Marks
Study_Hours     1.000000    1.000000     1.000000  0.999723
Attendance      1.000000    1.000000     1.000000  0.999723
Assignments     1.000000    1.000000     1.000000  0.999723
Marks           0.999723    0.999723     0.999723  1.000000
```

Heatmap

A correlation matrix can be visualized using a heatmap.

[13]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

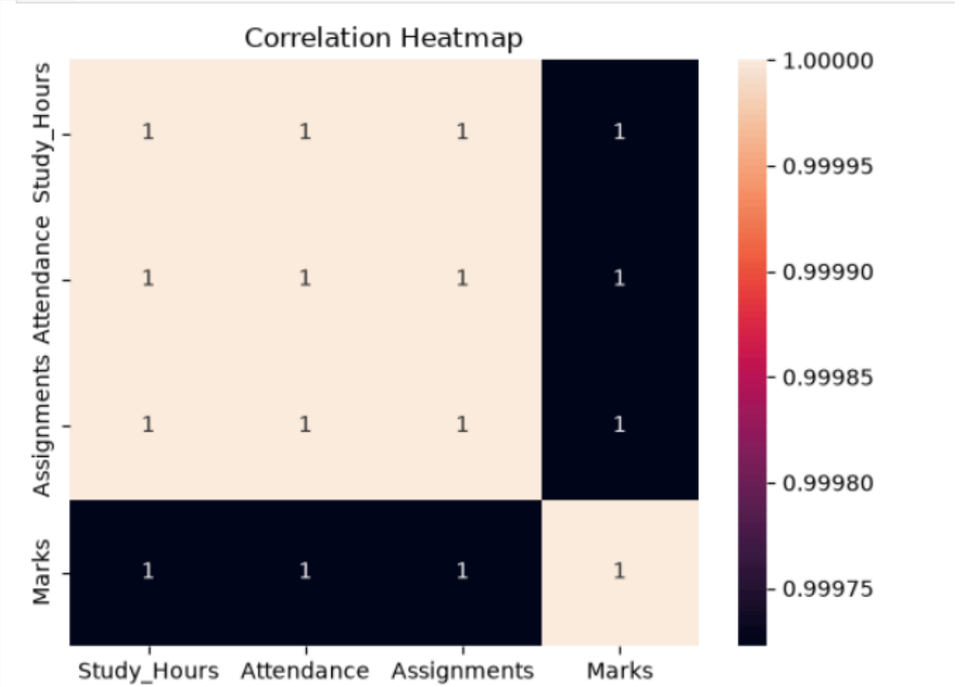
# Create the data
data = {
    "Study_Hours": [2, 3, 4, 5, 6, 7, 8],
    "Attendance": [65, 70, 75, 80, 85, 90, 95],
    "Assignments": [3, 4, 5, 6, 7, 8, 9],
    "Marks": [48, 55, 62, 70, 78, 85, 92]
}

# Create DataFrame
df = pd.DataFrame(data)

# Calculate correlation
correlation = df[
    ["Study_Hours", "Attendance", "Assignments", "Marks"]
].corr()

# Create heatmap
sns.heatmap(correlation, annot=True)

plt.title("Correlation Heatmap")
plt.show()
```



Look at the data:

Study Hours	Attendance	Assignments	Marks
2	65	3	48
3	70	4	55
4	75	5	62
5	80	6	70
6	85	7	78
7	90	8	85
8	95	9	92

Notice the pattern:

- Study Hours increases → Attendance always increases
- Study Hours increases → Assignments always increase
- Study Hours increases → Marks always increase

Everything is moving together in a **perfectly consistent pattern**.
Therefore, the correlation is approximately **+1**.

END OF MODULE - 02

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